

MOTION IN 1D

Grade 9 Physics · Your core study guide

How to use this guide

Look at what is actually happening.

Draw it before you write any symbols.

Work out the rule, and when you are allowed to use it.

Practise with help, then with less help, then on your own.

(shortened to fit this page)

Before you write a single formula

Draw or describe the motion.

Decide where zero is and which way counts as positive.

Write down what is happening at the start and at the end.

Name the thing the question is asking for.

Check the situation fits the rule you want to use.

Only then pick a formula.

(shortened to fit this page)

1.1 AVERAGE SPEED EQUAL DISTANCE

Average Speed Equal Distance

Picture what is happening before you reach for a formula.

Start with a real situation

A real situation before any symbols.

Each standard relation links a specific set of state variables and leaves exactly one out.

(shortened to fit this page)

The big idea

The one sentence worth remembering.

List what you know and what you want; pick the relation that contains exactly those and nothing you cannot get.

A relation is a statement about states; choosing
(shortened to fit this page)

When does this model work?

Check the situation fits before you use the rule.

State the relation, the variables it links, and the
(shortened to fit this page)

WHAT IS ACTUALLY HAPPENING

- 1 Read what physically happens, in order.
- 2 Mark the start state and the end state.
- 3 Mark the quantity the question asks for.



Link the real situation to a drawing before you calculate.

See it step by step

Draw what is happening before you write anything.

This skill declares no frame or sign obligation; state the system anyway before writing symbols.

Rebuild the relation from the meaning of the quantities rather than recalling its letters.

1. Mark the known and wanted state variables in the state table.
2. Find the relation whose variable set matches that marking.

(shortened to fit this page)

1.1 AVERAGE SPEED EQUAL DISTANCE

Average Speed Equal Distance

One question, solved all the way to a number.

Turn the Picture Into Symbols

Each mark in the drawing gets a name.

This skill declares no frame or sign obligation; state the system anyway before writing symbols.

(shortened to fit this page)

WHAT YOU KNOW AT THE START AND AT THE END

state variable	stated in the source	to find
distance	?	?
speed	?	?

Fill both instants first; the empty cell is what the question asks for.

The same situation drawn out. No new numbers are introduced here.

Watch one worked example

Every step, with the numbers.

A cyclist rides along a straight road. She covers the first half of the road at a steady 40 km/h and the second half at a steady 60 km/h. What is her average speed for the whole ride?

Set it up: you are following the cyclist, measuring from the start of the road, counting forward along the road as positive.

What you are told: speed on the first half = 40 km/h; speed on the second half = 60 km/h. What you want: average speed for the whole ride.

1. Set Up Your Directions: One straight road, no turning back, so the speeds are just how fast she is going forward. Forward along the road counts as positive. The road splits into two halves of equal length.

2. See It: Draw the road as two equal lengths side by side, one slow and one fast. Equal lengths, different speeds — so the slow half takes longer.

3. Which Rule Fits Here?: The speed is steady on each half, so time = distance / speed works on each half separately. Average speed always means total distance divided by total time — never the average of the two speeds. Equal distances at two steady speeds gives average speed = $2 \times v_1 \times v_2 / (v_1 + v_2)$.

4. Do the Working: The road length cancels out, which is why the answer does not depend on how long the road is. $v_{avg} = 2 \times v_1 \times v_2 / (v_1 + v_2) = 2 \times 40 \times 60 / (40 + 60) = 48 \text{ km/h}$

(shortened to fit this page)

AVERAGE SPEED EQUAL DISTANCE — watch it solved

Watch one worked example (continued)

6. Check Your Answer: The answer must land between the two speeds, and below their halfway value. Between the two speeds, and below the halfway value.

Answer: Her average speed for the whole ride is 48 km/h — closer to the slower half, because that is where she spends more of her time.

One thing to check: Your answer has to come out below the halfway value of the two speeds. If it came out exactly halfway, you averaged the speeds instead of the times.

Check it a different way: Pick any convenient road length — say 120 km. The first half is 60 km at 40 km/h, *(shortened to fit this page)*

Easy mistake to make

Lots of people fall for this one.

What lots of people think: Choosing a relation by how familiar it looks rather than by which variables it links.

Why that feels right: The shortcut looks sufficient because it reproduces the right answer in the cases seen so far, before the decisive physical feature is tested.

Now compare these two cases: A case pair in which the shortcut and the correct model give different answers.

Fix it like this: Predict both cases with the shortcut.

Fix it like this: Work both cases with the correct model.

Fix it like this: Say in one sentence which feature the shortcut ignores.

(shortened to fit this page)

1.1 AVERAGE SPEED EQUAL DISTANCE

Average Speed Equal Distance

Check it before you accept it.

Does the answer make sense?

Would this answer make sense in real life?

A check that can never come out wrong is not a check. A real one has to be able to tell you the answer is wrong.

Verify dimensions

(shortened to fit this page)

Quick check

Thirty seconds, no notes.

A cyclist rides along a straight road. She covers the first half of the road at a steady 24 km/h and the second half at a steady 36 km/h. What is her average speed for the whole ride?

Without looking back, say what your first move would be on the last question.

(shortened to fit this page)

Ready to practise?

You will meet it again here.

Where this idea turns up next: the same idea in a situation that looks nothing like this one. You will meet

(shortened to fit this page)

Try it with help

The first move is done for you.

A cyclist rides along a straight road. She covers the first half of the road at a steady 30 km/h and the second half at a steady 90 km/h. What is her average speed for the whole ride?

What you are told: speed on the first half = 30 km/h; speed on the second half = 90 km/h. What you want: average speed for the whole ride.

Use the supplied first-move cue and the pre-drawn frame; complete the remaining steps.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

1.1 AVERAGE SPEED EQUAL DISTANCE

Average Speed Equal Distance

Try it with help, then with less, then on your own.

Try with less help

One of the cues is gone now.

A cyclist rides along a straight road. She covers the first half of the road at a steady 20 km/h and the second half at a steady 30 km/h. What is her average speed for the whole ride?

What you are told: speed on the first half = 20 km/h; speed on the second half = 30 km/h. What you want: average speed for the whole ride.

One supplied cue has been removed; decide that step yourself and complete the rest.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

Now try it yourself

No cues this time. You have everything you need.

A cyclist rides along a straight road. She covers the first half of the road at a steady 60 km/h and the second half at a steady 45 km/h. What is her average speed for the whole ride?

What you are told: speed on the first half = 60 km/h; speed on the second half = 45 km/h. What you want: average speed for the whole ride.

No procedural cue is supplied beyond the situation and the required representation.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

1.1 AVERAGE SPEED EQUAL DISTANCE

Average Speed Equal Distance

Different pictures of the same idea.

CHECKS THAT COULD CATCH A MISTAKE

result under test

- ☐ verify dimensions
- ☐ verify total distance total time

Tick only after the check has actually been carried out on the result.

More ways to check this — Verification strip with one tick box per required second, different check. Represent

TWO CASES, ONE THING CHANGED

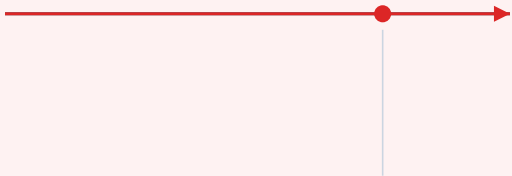
CASE A

the shortcut reproduces the right answer here



CASE B

the shortcut fails here



Decisive feature: A case pair in which the shortcut and the correct model give different answers.

More ways to check this — Two-case contrast panel in which exactly one physical feature differs. Representatio

1.2 DISTANCE DISPLACEMENT

Distance Displacement

Picture what is happening before you reach for a formula.

Start with a real situation

A real situation before any symbols.

Pick a fixed point to measure from and one direction to call positive, then keep both for the whole problem.

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The big idea

The one sentence worth remembering.

Decide once which way is positive; after that, a negative answer means the opposite direction, not a mistake.

A signed quantity has no meaning until the frame

(shortened to fit this page)

When does this model work?

Check the situation fits before you use the rule.

Every displacement, velocity and acceleration in

(shortened to fit this page)

WHAT IS ACTUALLY HAPPENING

- 1 Read what physically happens, in order.
- 2 Mark the start state and the end state.
- 3 Mark the quantity the question asks for.



Link the real situation to a drawing before you calculate.

See it step by step

Draw what is happening before you write anything.

State the origin, the positive direction, and which object the frame is attached to.

Draw the axis and the positive direction before substituting any number.

1. Choose and mark the origin.
2. Choose and mark the positive direction on the axis.

(shortened to fit this page)

1.2 DISTANCE DISPLACEMENT

Distance Displacement

One question, solved all the way to a number.

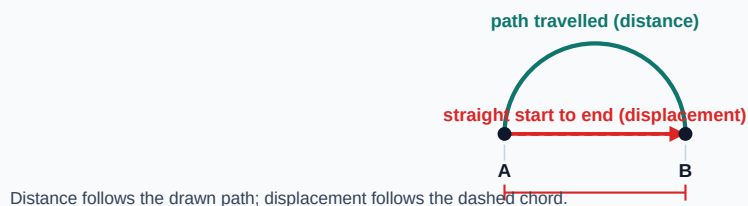
Turn the Picture Into Symbols

Each mark in the drawing gets a name.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

THE PATH YOU TRAVEL VERSUS THE STRAIGHT LINE



The same situation drawn out. No new numbers are introduced here.

Watch one worked example

Every step, with the numbers.

A runner starts at point A on a semicircular track of radius 20 m and runs along the curve all the way round to point B, directly opposite. How far has the runner travelled, and how far is she from where she started?

Set it up: you are following the runner, measuring from point A, where she starts, counting the straight line from A towards B as positive.

What you are told: radius of the track = 20 m. What you want: how far she ends up from the start.

1. Set Up Your Directions: Put the origin where she starts, so 'how far from the start' is measured from A. Start at A. B is directly opposite, across the circle.

2. See It: Draw the curved path she actually runs, then draw the straight arrow from A to B on top of it. They are two different lengths on the same picture. The curve is the path; the straight arrow is the gap between start and finish.

3. Which Rule Fits Here?: Distance follows the path her feet take. Displacement only cares about where she started and where she ended. Half a circle of radius R is $\pi \times R$ long; A to B straight across is $2 \times R$.

4. Do the Working: Half of a full circumference, since she runs half the circle. $s_{\text{path}} = \pi \times R = \pi \times 20 = 62.83 \text{ m}$

5. Do the Working: A to B straight across is the diameter, which is twice the radius. $s_{\text{disp}} = 2 \times R = 2 \times 20 =$
(shortened to fit this page)

DISTANCE DISPLACEMENT — watch it solved

Watch one worked example (continued)

6. What the Answer Means: She ran further than she ended up from the start, because the path bends. The bendier the path, the bigger the gap between the two numbers.

Answer: She runs 62.83 m of track, but she finishes only 40 m from where she started, in the straight line from A to B.

One thing to check: How far you end up from the start can never beat how far you ran. Check that your straight-line number is the smaller of the two.

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Easy mistake to make

Lots of people fall for this one.

What lots of people think: Treating downward motion as positive in one line and negative in the next.

Why that feels right: The shortcut looks sufficient because it reproduces the right answer in the cases seen so far, before the decisive physical feature is tested.

Now compare these two cases: A case pair in which the shortcut and the correct model give different answers.

Fix it like this: Predict both cases with the shortcut.

Fix it like this: Work both cases with the correct model.

Fix it like this: Say in one sentence which feature the shortcut ignores.

(shortened to fit this page)

1.2 DISTANCE DISPLACEMENT

Distance Displacement

Check it before you accept it.

Does the answer make sense?

Would this answer make sense in real life?

A check that can never come out wrong is not a check. A real one has to be able to tell you the answer is wrong.

Verify dimensions

(shortened to fit this page)

Quick check

Thirty seconds, no notes.

A runner starts at point A on a semicircular track of radius 28 m and runs along the curve all the way round to point B, directly opposite. How far has the runner travelled, and how far is she from where she started?

Without looking back, say what your first move would be on the last question.

(shortened to fit this page)

Ready to practise?

You will meet it again here.

Where this idea turns up next: the same idea in a situation that looks nothing like this one. You will meet

(shortened to fit this page)

Try it with help

The first move is done for you.

A runner starts at point A on a semicircular track of radius 35 m and runs along the curve all the way round to point B, directly opposite. How far has the runner travelled, and how far is she from where she started?

What you are told: radius of the track = 35 m. What you want: how far she ends up from the start.

Use the supplied first-move cue and the pre-drawn frame; complete the remaining steps.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

1.2 DISTANCE DISPLACEMENT

Distance Displacement

Try it with help, then with less, then on your own.

Try with less help

One of the cues is gone now.

A runner starts at point A on a semicircular track of radius 50 m and runs along the curve all the way round to point B, directly opposite. How far has the runner travelled, and how far is she from where she started?

What you are told: radius of the track = 50 m. What you want: how far she ends up from the start.

One supplied cue has been removed; decide that step yourself and complete the rest.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

Now try it yourself

No cues this time. You have everything you need.

A runner starts at point A on a semicircular track of radius 14 m and runs along the curve all the way round to point B, directly opposite. How far has the runner travelled, and how far is she from where she started?

What you are told: radius of the track = 14 m. What you want: how far she ends up from the start.

No procedural cue is supplied beyond the situation and the required representation.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

1.2 DISTANCE DISPLACEMENT

Distance Displacement

Different pictures of the same idea.

CHECKS THAT COULD CATCH A MISTAKE

result under test

- ☐ verify dimensions
- ☐ verify path vs displacement

Tick only after the check has actually been carried out on the result.

More ways to check this — Verification strip with one tick box per required second, different check. Represent

TWO CASES, ONE THING CHANGED

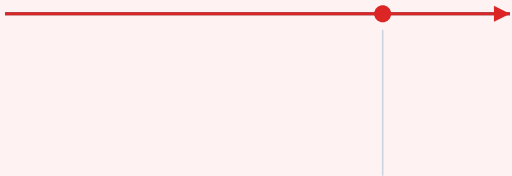
CASE A

the shortcut reproduces the right answer here



CASE B

the shortcut fails here



Decisive feature: A case pair in which the shortcut and the correct model give different answers.

More ways to check this — Two-case contrast panel in which exactly one physical feature differs. Representatio

1.3 FIRST EQUATION

First Equation

Picture what is happening before you reach for a formula.

Start with a real situation

A real situation before any symbols.

Ask what is being assumed constant, and whether the situation says it is.

(shortened to fit this page)

The big idea

The one sentence worth remembering.

Before using a relation, say in words what it assumes and point to where the question says that is true.

A relation is only valid while its assumptions

(shortened to fit this page)

When does this model work?

Check the situation fits before you use the rule.

Name the model, list its assumptions, and mark each assumption as stated, implied or absent.

(shortened to fit this page)

WHAT IS ACTUALLY HAPPENING

- 1 Read what physically happens, in order.
- 2 Mark the start state and the end state.
- 3 Mark the quantity the question asks for.



Link the real situation to a drawing before you calculate.

See it step by step

Draw what is happening before you write anything.

State the origin, the positive direction, and which object the frame is attached to.

Check each assumption of the candidate model against a stated condition.

1. List the candidate physical models for this situation.

2. Write each model's assumptions as a checklist.

(shortened to fit this page)

1.3 FIRST EQUATION

First Equation

One question, solved all the way to a number.

Turn the Picture Into Symbols

Each mark in the drawing gets a name.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

WHAT YOU KNOW AT THE START AND AT THE END

state variable	stated in the source	to find
acceleration	2 m/s^2	?
time	?	?

Fill both instants first; the empty cell is what the question asks for.

The same situation drawn out. No new numbers are introduced here.

Watch one worked example

Every step, with the numbers.

A skateboarder starts from rest on a flat straight path and speeds up steadily at 1.5 m/s^2. How long does it take her to reach 9 m/s?

Set it up: you are following the skateboarder, measuring from where she starts, counting the way she is heading as positive.

What you are told: starting velocity = 0 m/s; acceleration = 1.5 m/s^2; velocity she is aiming for = 9 m/s.
What you want: time taken.

1. Set Up Your Directions: One straight path, and she only goes one way, so the direction she is heading counts as positive. Forward is positive. She starts at rest at the origin.

2. See It: Speeding up steadily draws a straight sloping line on a velocity-time graph, starting at zero. A straight line from 0 up to the target velocity. The steepness is the acceleration.

3. Which Rule Fits Here?: The acceleration is stated as steady, so the constant-acceleration relations are allowed. If she were speeding up unevenly they would not be. Steady acceleration, so $v = u + a t$ applies over the whole interval.

4. Do the Working: Rearranging $v = u + a t$ for the time gives $t = (v - u) / a$. That relation has exactly the three things we know and the one we want. $t = (v-u)/a = (9-0)/1.5 = 6 \text{ s}$

(shortened to fit this page)

FIRST EQUATION — watch it solved

Watch one worked example (continued)

6. Check Your Answer: A speed divided by an acceleration leaves seconds, so the units are right. Units come out as seconds, and the time is positive.

Answer: She reaches 9 m/s after 6 s.

One thing to check: Divide a speed by an acceleration and what is left is seconds. If your answer came out in metres you picked a relation with a distance in it.

Check it a different way: Work out how far she travels, twice, by two different routes: once from the speeds alone using $(v^2 - u^2) / (2a)$, and once from the time you just found using $ut + \frac{1}{2}at^2$

(shortened to fit this page)

Easy mistake to make

Lots of people fall for this one.

What lots of people think: Using a constant-rate relation for a situation that changes rate part-way through.

Why that feels right: The shortcut looks sufficient because it reproduces the right answer in the cases seen so far, before the decisive physical feature is tested.

Now compare these two cases: A case pair in which the shortcut and the correct model give different answers.

Fix it like this: Predict both cases with the shortcut.

Fix it like this: Work both cases with the correct model.

Fix it like this: Say in one sentence which feature the shortcut ignores.

(shortened to fit this page)

1.3 FIRST EQUATION

First Equation

Check it before you accept it.

Does the answer make sense?

Would this answer make sense in real life?

A check that can never come out wrong is not a check. A real one has to be able to tell you the answer is wrong.

Verify dimensions

Check you were allowed to use that rule

(shortened to fit this page)

Quick check

Thirty seconds, no notes.

A skateboarder starts from rest on a flat straight path and speeds up steadily at 1.5 m/s^2 . How long does it take her to reach 6 m/s ?

Without looking back, say what your first move would be on the last question.

(shortened to fit this page)

Ready to practise?

You will meet it again here.

Where this idea turns up next: the same idea in a situation that looks nothing like this one. You will meet

(shortened to fit this page)

Try it with help

The first move is done for you.

A skateboarder starts from rest on a flat straight path and speeds up steadily at 2 m/s^2 . How long does it take her to reach 10 m/s ?

What you are told: starting velocity = 0 m/s ; acceleration = 2 m/s^2 ; velocity she is aiming for = 10 m/s .
What you want: time taken.

Use the supplied first-move cue and the pre-drawn frame; complete the remaining steps.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

1.3 FIRST EQUATION

First Equation

Try it with help, then with less, then on your own.

Try with less help

One of the cues is gone now.

A skateboarder starts from rest on a flat straight path and speeds up steadily at 2.5 m/s^2 . How long does it take her to reach 12 m/s ?

What you are told: starting velocity = 0 m/s ; acceleration = 2.5 m/s^2 ; velocity she is aiming for = 12 m/s .
What you want: time taken.

One supplied cue has been removed; decide that step yourself and complete the rest.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

Now try it yourself

No cues this time. You have everything you need.

A skateboarder starts from rest on a flat straight path and speeds up steadily at 3 m/s^2 . How long does it take her to reach 7.5 m/s ?

What you are told: starting velocity = 0 m/s ; acceleration = 3 m/s^2 ; velocity she is aiming for = 7.5 m/s .
What you want: time taken.

No procedural cue is supplied beyond the situation and the required representation.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

1.3 FIRST EQUATION

First Equation

Different pictures of the same idea.

WHERE ZERO IS AND WHICH WAY IS POSITIVE



Declared positive direction: declared +. Re-sign every given quantity against it here.

More ways to see this — Signed axis strip marking the origin, the declared positive direction, and each signed

WHAT EACH LETTER MEANS IN THE PICTURE

DIAGRAM FEATURE



SYMBOL IN THE RELATION

- u** arrow at the start instant
- v** arrow at the end instant
- a** change of the arrow between the instants

Bind every symbol to a diagram feature before substituting.

More ways to check this — Bridge panel linking each symbol of the relation to the diagram feature it names. Re

WHEN YOU ARE ALLOWED TO USE THIS RULE

model: phy model uniform acceleration

☐ acceleration is constant over the interval stated / implied / absent

If any condition is absent the relation below this line may not be used.

More ways to check this — Whether the rule is allowed here gate listing each assumption with its stated, impli

1.3 FIRST EQUATION

First Equation

Different pictures of the same idea.

CHECKS THAT COULD CATCH A MISTAKE

result under test

- ☐ verify dimensions
- ☐ check you were allowed to use that rule
- ☐ verify velocity endpoint

Tick only after the check has actually been carried out on the result.

More ways to check this — Verification strip with one tick box per required second, different check. Represent

TWO CASES, ONE THING CHANGED

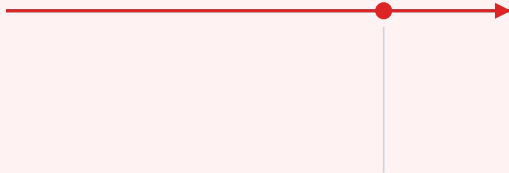
CASE A

the shortcut reproduces the right answer here



CASE B

the shortcut fails here



Decisive feature: A case pair in which the shortcut and the correct model give different answers.

More ways to check this — Two-case contrast panel in which exactly one physical feature differs. Representatio

1.4 FRAME SIGN SETUP

Frame Sign Setup

Picture what is happening before you reach for a formula.

Start with a real situation

A real situation before any symbols.

A graph is a record of the motion, not a picture of the path.

(shortened to fit this page)

The big idea

The one sentence worth remembering.

Ask first: do I need how fast this is changing, or how much it has changed in total?

On a graph of a state variable against time, slope is a rate and signed area is a change; which one

(shortened to fit this page)

When does this model work?

Check the situation fits before you use the rule.

Name the model, list its assumptions, and mark each assumption as stated, implied or absent.

acceleration is constant over the interval

acceleration is constant within each phase

(shortened to fit this page)

WHAT IS ACTUALLY HAPPENING

- 1 Read what physically happens, in order.
- 2 Mark the start state and the end state.
- 3 Mark the quantity the question asks for.



Link the real situation to a drawing before you calculate.

See it step by step

Draw what is happening before you write anything.

State the origin, the positive direction, and which object the frame is attached to.

Decode slope and area separately and say what each one is physically.

1. Read and state both axis labels and units.

2. Decide whether the question asks for a rate or an accumulated change.

(shortened to fit this page)

1.4 FRAME SIGN SETUP

Frame Sign Setup

One question, solved all the way to a number.

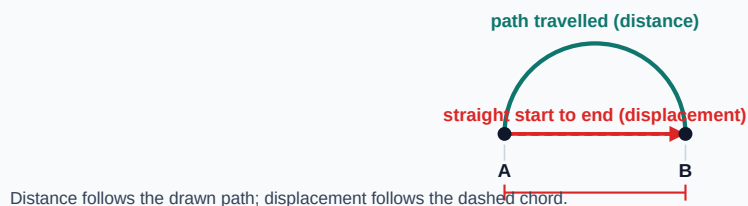
Turn the Picture Into Symbols

Each mark in the drawing gets a name.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

THE PATH YOU TRAVEL VERSUS THE STRAIGHT LINE



The same situation drawn out. No new numbers are introduced here.

Watch one worked example

Every step, with the numbers.

A skateboarder starts from rest on a flat straight path and speeds up steadily at 1.5 m/s^2 . How long does it take her to reach 9 m/s ?

Set it up: you are following the skateboarder, measuring from where she starts, counting the way she is heading as positive.

What you are told: starting velocity = 0 m/s ; acceleration = 1.5 m/s^2 ; velocity she is aiming for = 9 m/s .
What you want: time taken.

1. Set Up Your Directions: One straight path, and she only goes one way, so the direction she is heading counts as positive. Forward is positive. She starts at rest at the origin.

2. See It: Speeding up steadily draws a straight sloping line on a velocity-time graph, starting at zero. A straight line from 0 up to the target velocity. The steepness is the acceleration.

3. Which Rule Fits Here?: The acceleration is stated as steady, so the constant-acceleration relations are allowed. If she were speeding up unevenly they would not be. Steady acceleration, so $v = u + a t$ applies over the whole interval.

4. Do the Working: Rearranging $v = u + a t$ for the time gives $t = (v - u) / a$. That relation has exactly the three things we know and the one we want. $t = (v - u) / a = (9 - 0) / 1.5 = 6 \text{ s}$

(shortened to fit this page)

FRAME SIGN SETUP — watch it solved

Watch one worked example (continued)

6. Check Your Answer: A speed divided by an acceleration leaves seconds, so the units are right. Units come out as seconds, and the time is positive.

Answer: She reaches 9 m/s after 6 s.

One thing to check: Divide a speed by an acceleration and what is left is seconds. If your answer came out in metres you picked a relation with a distance in it.

Check it a different way: Work out how far she travels, twice, by two different routes: once from the speeds alone using $(v^2 - u^2) / (2a)$, and once from the time you just found using $ut + \frac{1}{2}at^2$

(shortened to fit this page)

Easy mistake to make

Lots of people fall for this one.

What lots of people think: Reading a velocity-time graph as if it were a picture of the path travelled.

Why that feels right: The shortcut looks sufficient because it reproduces the right answer in the cases seen so far, before the decisive physical feature is tested.

Now compare these two cases: A case pair in which the shortcut and the correct model give different answers.

Fix it like this: Predict both cases with the shortcut.

Fix it like this: Work both cases with the correct model.

Fix it like this: Say in one sentence which feature the shortcut ignores.

(shortened to fit this page)

1.4 FRAME SIGN SETUP

Frame Sign Setup

Check it before you accept it.

Does the answer make sense?

Would this answer make sense in real life?

A check that can never come out wrong is not a check. A real one has to be able to tell you the answer is wrong.

Verify acceleration sign units

Verify constant g assumptions

(shortened to fit this page)

Quick check

Thirty seconds, no notes.

A skateboarder starts from rest on a flat straight path and speeds up steadily at 1.5 m/s^2 . How long does it take her to reach 6 m/s ?

Without looking back, say what your first move would be on the last question.

(shortened to fit this page)

Ready to practise?

You will meet it again here.

Where this idea turns up next: the same idea in a situation that looks nothing like this one. You will meet

(shortened to fit this page)

1.4 FRAME SIGN SETUP

Frame Sign Setup

Try it with help, then with less, then on your own.

Try it with help

The first move is done for you.

A skateboarder starts from rest on a flat straight path and speeds up steadily at 2 m/s^2 . How long does it take her to reach 10 m/s ?

What you are told: starting velocity = 0 m/s ; acceleration = 2 m/s^2 ; velocity she is aiming for = 10 m/s .

What you want: time taken.

Use the supplied first-move cue and the pre-drawn frame; complete the remaining steps.

The full answer is in the answers appendix — try it first.

Once you have an answer, check it: Verify acceleration sign units; Verify constant g assumptions; Verify dimensions; Verify graph slope sign; Verify interval not cumulative; Verify maximum height endpoint;

(shortened to fit this page)

Try with less help

One of the cues is gone now.

A skateboarder starts from rest on a flat straight path and speeds up steadily at 2.5 m/s^2 . How long does it take her to reach 12 m/s ?

What you are told: starting velocity = 0 m/s ; acceleration = 2.5 m/s^2 ; velocity she is aiming for = 12 m/s .

What you want: time taken.

One supplied cue has been removed; decide that step yourself and complete the rest.

The full answer is in the answers appendix — try it first.

Once you have an answer, check it: Verify acceleration sign units; Verify constant g assumptions; Verify dimensions; Verify graph slope sign; Verify interval not cumulative; Verify maximum height endpoint;

(shortened to fit this page)

1.4 FRAME SIGN SETUP

Frame Sign Setup

Try it with help, then with less, then on your own.

Now try it yourself

No cues this time. You have everything you need.

A skateboarder starts from rest on a flat straight path and speeds up steadily at 3 m/s^2 . How long does it take her to reach 7.5 m/s ?

What you are told: starting velocity = 0 m/s ; acceleration = 3 m/s^2 ; velocity she is aiming for = 7.5 m/s .

What you want: time taken.

No procedural cue is supplied beyond the situation and the required representation.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

WHERE ZERO IS AND WHICH WAY IS POSITIVE



Declared positive direction: right. Re-sign every given quantity against it here.

More ways to see this — Signed axis strip marking the origin, the declared positive direction, and each signed

WHERE IT IS AT EQUAL TIME STEPS

Gaps are equal time steps; unequal gaps mean the speed is changing.



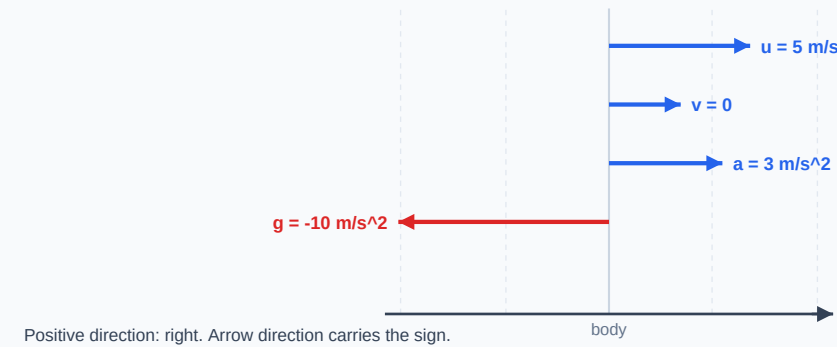
More ways to see this — Strip of successive body positions at equal time intervals along the declared axis. Re

1.4 FRAME SIGN SETUP

Frame Sign Setup

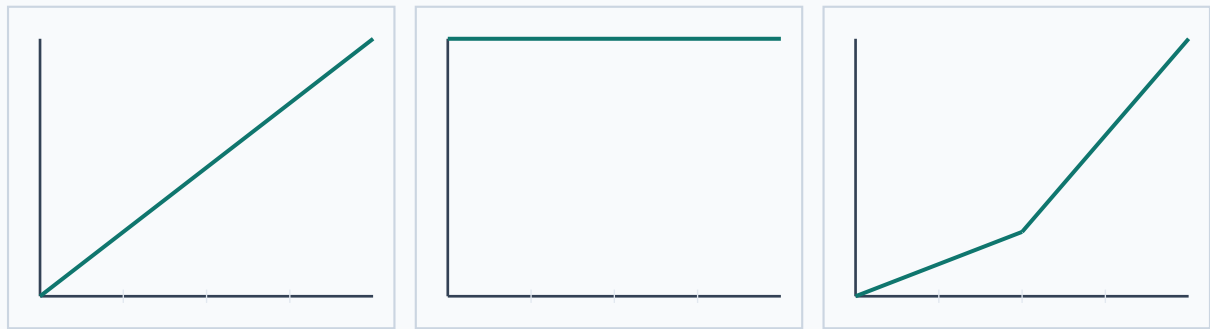
Different pictures of the same idea.

WHICH WAY EACH ARROW POINTS



More ways to draw this — Vector state view with one labelled arrow per directed quantity at the named instant.

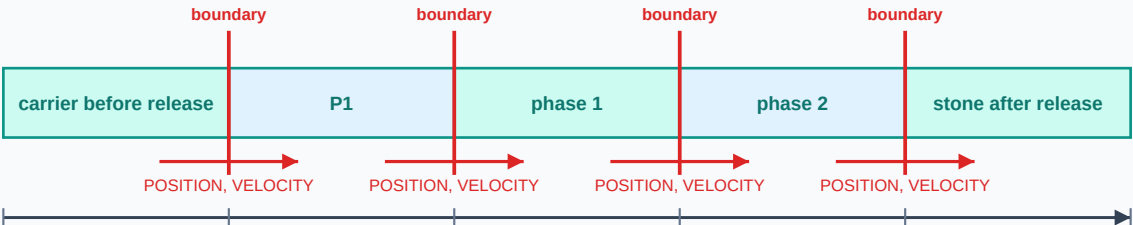
THE CHOICES, DRAWN ON THE SAME AXES



Reject each option by naming the feature it gets wrong.

More ways to draw this — Set of candidate graphs drawn on identical axes so one feature separates them. Repres

WHAT CARRIES OVER WHEN THE MOTION CHANGES



The end state of one phase is the start state of the next; only the rate may jump.

More ways to draw this — Phase strip with boundary markers and the continuity state labelled across each bound

1.4 FRAME SIGN SETUP

Frame Sign Setup

Different pictures of the same idea.

POSITION AGAINST TIME: STEEPNESS IS SPEED



More ways to draw this — Position-time graph with labelled axes and a rise-over-run slope triangle. Representa

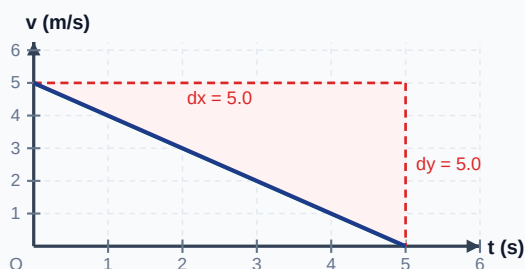
MEASURED FROM WHICH OBJECT?



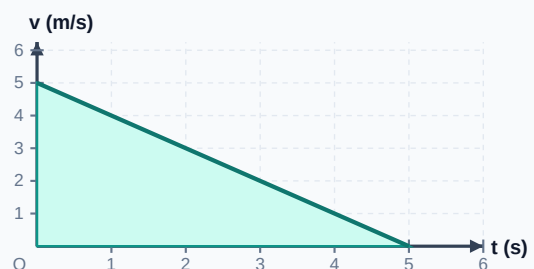
More ways to draw this — Relative-frame view marking the frame body and the relative quantity between the two

STEEPNESS OR AREA? ONE GRAPH, TWO READINGS

READ AS SLOPE = a rate



READ AS SIGNED AREA = a change



More ways to draw this — Decoder showing the same curve read once as a slope and once as a signed area. Repres

1.4 FRAME SIGN SETUP

Frame Sign Setup

Different pictures of the same idea.

WHAT YOU KNOW AT THE START AND AT THE END

state variable	stated in the source	to find
acceleration	3 m/s ²	?
displacement	?	?
distance	?	?
height	45 m	?
position	?	?
relative acceleration	?	?
time	5 s	?

Fill both instants first; the empty cell is what the question asks for.

More ways to draw this — Ruled state table with a start row and an end row and the unknown cell marked. Repres

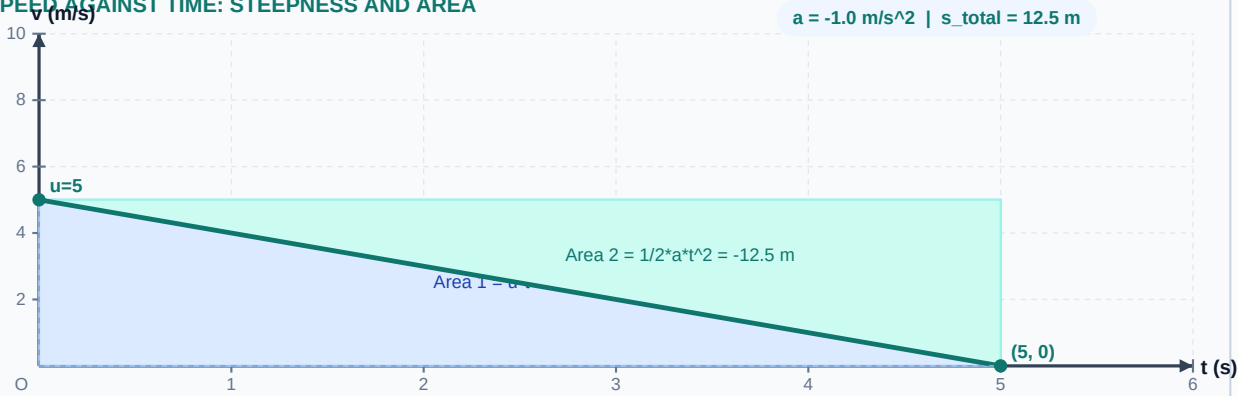
ONE NAMED SECOND IS A DIFFERENCE



Compute the quantity at each bounding instant, then subtract.

More ways to draw this — Timeline marking the bounding instants of the named interval and the span between the

SPEED AGAINST TIME: STEEPNESS AND AREA



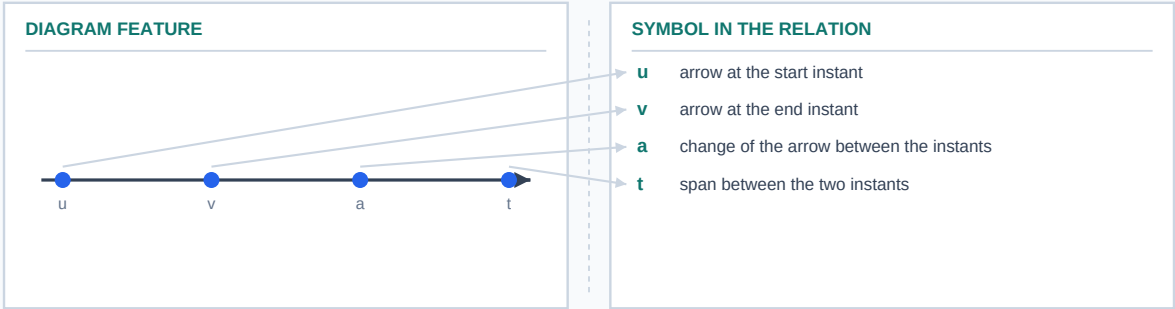
More ways to draw this — Velocity-time graph with labelled axes and shaded signed area under the line. Represe

1.4 FRAME SIGN SETUP

Frame Sign Setup

Different pictures of the same idea.

WHAT EACH LETTER MEANS IN THE PICTURE



Bind every symbol to a diagram feature before substituting.

More ways to check this — Bridge panel linking each symbol of the relation to the diagram feature it names. Re

WHEN YOU ARE ALLOWED TO USE THIS RULE

model: phy model constant g 1d, phy model graph kinematics, phy model moving release,

- | | |
|--|---------------------------|
| ☐ acceleration is constant over the interval | stated / implied / absent |
| ☐ acceleration is constant within each phase | stated / implied / absent |
| ☐ acceleration is constant within each stated phase | stated / implied / absent |
| ☐ acceleration is uniform over the 5 s interval, or the requested quantity is | stated / implied / absent |
| ☐ air resistance neglected | stated / implied / absent |

If any condition is absent the relation below this line may not be used.

More ways to check this — Whether the rule is allowed here gate listing each assumption with its stated, impli

CHECKS THAT COULD CATCH A MISTAKE

result under test

- ☐ verify acceleration sign units
- ☐ verify constant g assumptions
- ☐ verify dimensions
- ☐ verify graph slope sign
- ☐ verify interval not cumulative

Tick only after the check has actually been carried out on the result.

More ways to check this — Verification strip with one tick box per required second, different check. Represent

1.4 FRAME SIGN SETUP

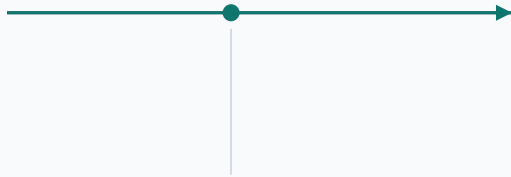
Frame Sign Setup

Different pictures of the same idea.

TWO CASES, ONE THING CHANGED

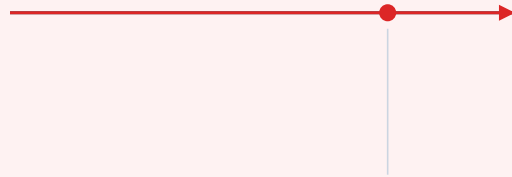
CASE A

the shortcut reproduces the right answer here



CASE B

the shortcut fails here



Decisive feature: A case pair in which the shortcut and the correct model give different answers.

More ways to check this — Two-case contrast panel in which exactly one physical feature differs. Representatio

1.5 FREE FALL CONSTANT G

Free Fall Constant G

Picture what is happening before you reach for a formula.

Start with a real situation

A real situation before any symbols.

The object does not teleport: position and velocity at the instant of the change belong to both phases.

(shortened to fit this page)

The big idea

The one sentence worth remembering.

Solve one stage at a time, and hand the end of each stage to the start of the next.

At a phase boundary the state is continuous

(shortened to fit this page)

When does this model work?

Check the situation fits before you use the rule.

Name the model, list its assumptions, and mark each assumption as stated, implied or absent.

air resistance neglected

g constant

(shortened to fit this page)

WHAT IS ACTUALLY HAPPENING

- 1 Read what physically happens, in order.
- 2 Mark the start state and the end state.
- 3 Mark the quantity the question asks for.



Link the real situation to a drawing before you calculate.

See it step by step

Draw what is happening before you write anything.

State the origin, the positive direction, and which object the frame is attached to.

Draw a phase strip and mark the quantity that must carry over unchanged.

1. Mark every instant at which the motion changes character.
2. Give each phase its own state table and relation.

(shortened to fit this page)

1.5 FREE FALL CONSTANT G

Free Fall Constant G

One question, solved all the way to a number.

Turn the Picture Into Symbols

Each mark in the drawing gets a name.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

WHICH WAY EACH ARROW POINTS



The same situation drawn out. No new numbers are introduced here.

Watch one worked example

Every step, with the numbers.

A small stone is dropped — not thrown — from a bridge 45 m above the water. Take gravity as 10 m/s^2 and ignore air resistance. How long does the stone take to reach the water, and how fast is it moving when it gets there?

Set it up: you are following the stone, measuring from the point it is released from, counting downwards, the way it is about to move as positive.

What you are told: height of the drop = 45 m; acceleration due to gravity = 10 m/s^2 ; starting velocity = 0 m/s. What you want: time to reach the water.

1. Set Up Your Directions: Everything in this problem happens downwards, so calling downwards positive keeps every number positive. Calling upwards positive also works — you just carry minus signs the whole way. Down is positive. Zero is the release point.

2. See It: Write the start state and the end state in a table: at the start, speed 0 and position 0; at the end, position h and speed unknown. Start: 0 m, 0 m/s. End: h down, speed unknown. Time is what we want.

3. Which Rule Fits Here?: With air resistance ignored, gravity alone acts and the acceleration stays steady, so the constant-acceleration relations are allowed. For a light object like a feather they would not be. Free fall: steady acceleration g downwards, no other force.

4. Do the Working: From $h = \frac{1}{2} g t^2$ with no starting speed, the time is the square root of $2h/g$.

(shortened to fit this page)

FREE FALL CONSTANT G — watch it solved

Watch one worked example (continued)

5. Do the Working: It starts at rest and gains g every second, so its speed on arrival is g times the fall time.
 $v_{\text{land}} = g \cdot t = 10 \cdot 3 = 30 \text{ m/s}$

6. What the Answer Means: Both numbers are positive because both point downwards, the direction we chose as positive. A few seconds of fall, arriving fast.

Answer: The stone reaches the water after 3 s, moving at 30 m/s downwards.

One thing to check: Doubling the drop height does not double the time — the time follows the square root of the height. So a drop four times as high takes only twice as long.

Check it a different way: Put your time back into the distance relation and see whether it rebuilds the drop:
(shortened to fit this page)

Easy mistake to make

Lots of people fall for this one.

What lots of people think: Applying a single whole-journey relation across a change of acceleration.

Why that feels right: The shortcut looks sufficient because it reproduces the right answer in the cases seen so far, before the decisive physical feature is tested.

Now compare these two cases: A case pair in which the shortcut and the correct model give different answers.

Fix it like this: Predict both cases with the shortcut.

Fix it like this: Work both cases with the correct model.

Fix it like this: Say in one sentence which feature the shortcut ignores.

(shortened to fit this page)

1.5 FREE FALL CONSTANT G

Free Fall Constant G

Check it before you accept it.

Does the answer make sense?

Would this answer make sense in real life?

A check that can never come out wrong is not a check. A real one has to be able to tell you the answer is wrong.

Verify constant g assumptions

Verify dimensions

(shortened to fit this page)

Quick check

Thirty seconds, no notes.

A small stone is dropped — not thrown — from a bridge 5 m above the water. Take gravity as 10 m/s^2 and ignore air resistance. How long does the stone take to reach the water, and how fast is it moving when it gets there?

Without looking back, say what your first move would be on the last question.

(shortened to fit this page)

Ready to practise?

You will meet it again here.

Where this idea turns up next: the same idea in a situation that looks nothing like this one. You will meet

(shortened to fit this page)

1.5 FREE FALL CONSTANT G

Free Fall Constant G

Try it with help, then with less, then on your own.

Try it with help

The first move is done for you.

A small stone is dropped — not thrown — from a bridge 20 m above the water. Take gravity as 10 m/s^2 and ignore air resistance. How long does the stone take to reach the water, and how fast is it moving when it gets there?

What you are told: height of the drop = 20 m; acceleration due to gravity = 10 m/s^2 ; starting velocity = 0 m/s. What you want: time to reach the water.

Use the supplied first-move cue and the pre-drawn frame; complete the remaining steps.

The full answer is in the answers appendix — try it first.

Once you have an answer, check it: Verify constant g assumptions; Verify dimensions; Verify maximum

(shortened to fit this page)

Try with less help

One of the cues is gone now.

A small stone is dropped — not thrown — from a bridge 80 m above the water. Take gravity as 10 m/s^2 and ignore air resistance. How long does the stone take to reach the water, and how fast is it moving when it gets there?

What you are told: height of the drop = 80 m; acceleration due to gravity = 10 m/s^2 ; starting velocity = 0 m/s. What you want: time to reach the water.

One supplied cue has been removed; decide that step yourself and complete the rest.

The full answer is in the answers appendix — try it first.

Once you have an answer, check it: Verify constant g assumptions; Verify dimensions; Verify maximum

(shortened to fit this page)

1.5 FREE FALL CONSTANT G

Free Fall Constant G

Try it with help, then with less, then on your own.

Now try it yourself
No cues this time. You have everything you need.

A small stone is dropped — not thrown — from a bridge 125 m above the water. Take gravity as 10 m/s² and ignore air resistance. How long does the stone take to reach the water, and how fast is it moving when it gets there?

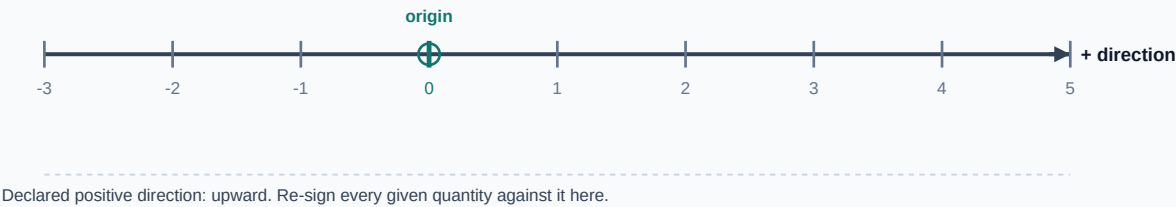
What you are told: height of the drop = 125 m; acceleration due to gravity = 10 m/s²; starting velocity = 0 m/s. What you want: time to reach the water.

No procedural cue is supplied beyond the situation and the required representation.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

WHERE ZERO IS AND WHICH WAY IS POSITIVE



More ways to see this — Signed axis strip marking the origin, the declared positive direction, and each signed

WHERE IT IS AT EQUAL TIME STEPS

Gaps are equal time steps; unequal gaps mean the speed is changing.



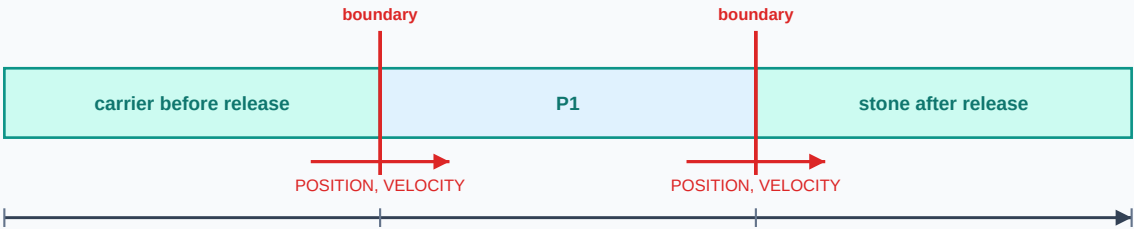
More ways to see this — Strip of successive body positions at equal time intervals along the declared axis. Re

1.5 FREE FALL CONSTANT G

Free Fall Constant G

Different pictures of the same idea.

WHAT CARRIES OVER WHEN THE MOTION CHANGES



The end state of one phase is the start state of the next; only the rate may jump.

More ways to draw this — Phase strip with boundary markers and the continuity state labelled across each bound

WHAT YOU KNOW AT THE START AND AT THE END

state variable	stated in the source	to find
acceleration	?	?
displacement	?	?
height	45 m	?
position	?	?
time	?	?
velocity	8 m/s	?

Fill both instants first; the empty cell is what the question asks for.

More ways to draw this — Ruled state table with a start row and an end row and the unknown cell marked. Repres

WHAT EACH LETTER MEANS IN THE PICTURE

DIAGRAM FEATURE

SYMBOL IN THE RELATION

u

arrow at the start instant

v

arrow at the end instant

Bind every symbol to a diagram feature before substituting.

More ways to check this — Bridge panel linking each symbol of the relation to the diagram feature it names. Re

1.5 FREE FALL CONSTANT G

Free Fall Constant G

Different pictures of the same idea.

WHEN YOU ARE ALLOWED TO USE THIS RULE

model: phy model constant g 1d, phy model moving release

☐ air resistance neglected	stated / implied / absent
☐ g constant	stated / implied / absent
☐ g constant after release	stated / implied / absent
☐ released body inherits carrier velocity at release	stated / implied / absent

If any condition is absent the relation below this line may not be used.

More ways to check this — Whether the rule is allowed here gate listing each assumption with its stated, impli

CHECKS THAT COULD CATCH A MISTAKE

result under test

- ☐ ▶ verify constant g assumptions
- ☐ ▶ verify dimensions
- ☐ ▶ verify maximum height endpoint
- ☐ ▶ check you were allowed to use that rule
- ☐ ▶ verify release state continuity

Tick only after the check has actually been carried out on the result.

More ways to check this — Verification strip with one tick box per required second, different check. Represent

TWO CASES, ONE THING CHANGED

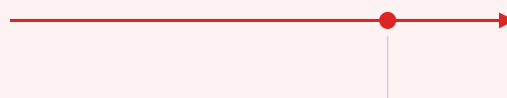
CASE A

the shortcut reproduces the right answer her
e



CASE B

the shortcut fails here



Decisive feature: A case pair in which the shortcut and the correct model give different answers.

More ways to check this — Two-case contrast panel in which exactly one physical feature differs. Representatio

1.6 GRAPH ACCELERATION

Graph Acceleration

Picture what is happening before you reach for a formula.

Start with a real situation

A real situation before any symbols.

A graph is a record of the motion, not a picture of the path.

(shortened to fit this page)

The big idea

The one sentence worth remembering.

Ask first: do I need how fast this is changing, or how much it has changed in total?

On a graph of a state variable against time, slope is a rate and signed area is a change; which one

(shortened to fit this page)

When does this model work?

Check the situation fits before you use the rule.

Slope gives the rate of the plotted quantity;

(shortened to fit this page)

WHAT IS ACTUALLY HAPPENING

- 1 Read what physically happens, in order.
- 2 Mark the start state and the end state.
- 3 Mark the quantity the question asks for.



Link the real situation to a drawing before you calculate.

See it step by step

Draw what is happening before you write anything.

State the origin, the positive direction, and which object the frame is attached to.

Decode slope and area separately and say what each one is physically.

1. Read and state both axis labels and units.
2. Decide whether the question asks for a rate or an accumulated change.

(shortened to fit this page)

1.6 GRAPH ACCELERATION

Graph Acceleration

One question, solved all the way to a number.

Turn the Picture Into Symbols

Each mark in the drawing gets a name.

State the origin, the positive direction, and which object the frame is attached to.

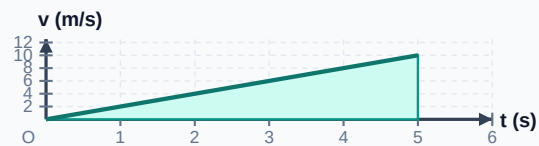
(shortened to fit this page)

STEEPNESS OR AREA? ONE GRAPH, TWO READINGS

READ AS SLOPE = a rate



READ AS SIGNED AREA = a change



The same situation drawn out. No new numbers are introduced here.

Watch one worked example

Every step, with the numbers.

A velocity-time graph for a trolley is a straight line. At 0 s the line reads 2 m/s, and at 4 s it reads 14 m/s. What is the trolley's acceleration?

Set it up: you are following the trolley, measuring from the start of the timing, counting the direction the graph calls positive velocity as positive.

What you are told: velocity at the first instant = 2 m/s; first instant = 0 s; velocity at the second instant = 14 m/s; second instant = 4 s. What you want: acceleration.

1. Set Up Your Directions: Read both axis labels first. The up axis is velocity, the across axis is time — so the steepness has velocity on top and time underneath. Up axis: velocity in m/s. Across axis: time in s.
2. See It: Mark the two points on the line and draw the step between them: across for the time, up for the velocity gained. A right-angled step under the line: across is the time taken, up is the velocity gained.
3. Which Rule Fits Here?: The line is straight, so the velocity changes at the same rate the whole way. That is exactly what constant acceleration means. Straight line on a velocity-time graph means steady acceleration, and the steepness is that acceleration.
4. Do the Working: Steepness means change in the up quantity divided by change in the across quantity.

$$a = (v_2 - v_1) / (t_2 - t_1) = (14 - 2) / (4 - 0) = 3 \text{ m/s}^2$$

(shortened to fit this page)

GRAPH ACCELERATION — watch it solved

Watch one worked example (continued)

direction it is already going. It gains that much velocity every second.

6. Check Your Answer: Metres per second, divided by seconds, is metres per second squared — the units of an acceleration. The units come out as m/s^2 .

Answer: The trolley's acceleration is 3 m/s^2 — it gains that much velocity every second.

One thing to check: The steepness of a velocity-time line is an acceleration, in m/s^2 . If your answer came out in m/s you read the height of the line instead of how steeply it climbs.

Check it a different way: Run it backwards through the other graph. On an acceleration-time graph, the
(shortened to fit this page)

Easy mistake to make

Lots of people fall for this one.

What lots of people think: Reading a velocity-time graph as if it were a picture of the path travelled.

Why that feels right: The shortcut looks sufficient because it reproduces the right answer in the cases seen so far, before the decisive physical feature is tested.

Now compare these two cases: A case pair in which the shortcut and the correct model give different answers.

Fix it like this: Predict both cases with the shortcut.

Fix it like this: Work both cases with the correct model.

Fix it like this: Say in one sentence which feature the shortcut ignores.

(shortened to fit this page)

1.6 GRAPH ACCELERATION

Graph Acceleration

Check it before you accept it.

Does the answer make sense?

Would this answer make sense in real life?

A check that can never come out wrong is not a check. A real one has to be able to tell you the answer is wrong.

(shortened to fit this page)

Quick check

Thirty seconds, no notes.

A velocity-time graph for a trolley is a straight line. At 1 s the line reads 3 m/s, and at 4 s it reads 16 m/s. What is the trolley's acceleration?

Without looking back, say what your first move would be on the last question.

(shortened to fit this page)

Ready to practise?

You will meet it again here.

Where this idea turns up next: the same idea in a situation that looks nothing like this one. You will meet

(shortened to fit this page)

Try it with help

The first move is done for you.

A velocity-time graph for a trolley is a straight line. At 1 s the line reads 0 m/s, and at 5 s it reads 12 m/s. What is the trolley's acceleration?

What you are told: velocity at the first instant = 0 m/s; first instant = 1 s; velocity at the second instant = 12 m/s; second instant = 5 s. What you want: acceleration.

Use the supplied first-move cue and the pre-drawn frame; complete the remaining steps.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

1.6 GRAPH ACCELERATION

Graph Acceleration

Try it with help, then with less, then on your own.

Try with less help

One of the cues is gone now.

A velocity-time graph for a trolley is a straight line. At 2 s the line reads 4 m/s, and at 6 s it reads 10 m/s. What is the trolley's acceleration?

What you are told: velocity at the first instant = 4 m/s; first instant = 2 s; velocity at the second instant = 10 m/s; second instant = 6 s. What you want: acceleration.

One supplied cue has been removed; decide that step yourself and complete the rest.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

Now try it yourself

No cues this time. You have everything you need.

A velocity-time graph for a trolley is a straight line. At 0 s the line reads 1 m/s, and at 8 s it reads 9 m/s. What is the trolley's acceleration?

What you are told: velocity at the first instant = 1 m/s; first instant = 0 s; velocity at the second instant = 9 m/s; second instant = 8 s. What you want: acceleration.

No procedural cue is supplied beyond the situation and the required representation.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

1.6 GRAPH ACCELERATION

Graph Acceleration

Different pictures of the same idea.

CHECKS THAT COULD CATCH A MISTAKE

result under test

☐ verify source representation present

Tick only after the check has actually been carried out on the result.

More ways to check this — Verification strip with one tick box per required second, different check. Represent

TWO CASES, ONE THING CHANGED

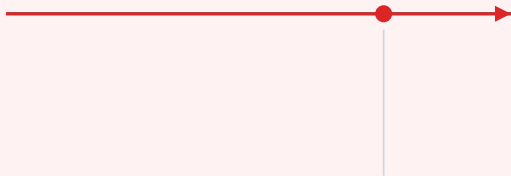
CASE A

the shortcut reproduces the right answer here



CASE B

the shortcut fails here



Decisive feature: A case pair in which the shortcut and the correct model give different answers.

More ways to check this — Two-case contrast panel in which exactly one physical feature differs. Representatio

1.7 Checking a rule fits the situation

Checking a rule fits the situation

Picture what is happening before you reach for a formula.

Start with a real situation

A real situation before any symbols.

The object does not teleport: position and velocity at the instant of the change belong to both phases.

(shortened to fit this page)

The big idea

The one sentence worth remembering.

Solve one stage at a time, and hand the end of each stage to the start of the next.

At a phase boundary the state is continuous

(shortened to fit this page)

When does this model work?

Check the situation fits before you use the rule.

Name the model, list its assumptions, and mark each assumption as stated, implied or absent.

acceleration is constant over the interval

acceleration is constant within each phase

(shortened to fit this page)

WHAT IS ACTUALLY HAPPENING

- 1 Read what physically happens, in order.
- 2 Mark the start state and the end state.
- 3 Mark the quantity the question asks for.



Link the real situation to a drawing before you calculate.

See it step by step

Draw what is happening before you write anything.

State the origin, the positive direction, and which object the frame is attached to.

Draw a phase strip and mark the quantity that must carry over unchanged.

1. Mark every instant at which the motion changes character.
2. Give each phase its own state table and relation.

(shortened to fit this page)

1.7 Checking a rule fits the situation

Checking a rule fits the situation

One question, solved all the way to a number.

Turn the Picture Into Symbols

Each mark in the drawing gets a name.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

WHICH WAY EACH ARROW POINTS



The same situation drawn out. No new numbers are introduced here.

Watch one worked example

Every step, with the numbers.

A skateboarder starts from rest on a flat straight path and speeds up steadily at 1.5 m/s^2 . How long does it take her to reach 9 m/s ?

Set it up: you are following the skateboarder, measuring from where she starts, counting the way she is heading as positive.

What you are told: starting velocity = 0 m/s ; acceleration = 1.5 m/s^2 ; velocity she is aiming for = 9 m/s .
What you want: time taken.

1. Set Up Your Directions: One straight path, and she only goes one way, so the direction she is heading counts as positive. Forward is positive. She starts at rest at the origin.

2. See It: Speeding up steadily draws a straight sloping line on a velocity-time graph, starting at zero. A straight line from 0 up to the target velocity. The steepness is the acceleration.

3. Which Rule Fits Here?: The acceleration is stated as steady, so the constant-acceleration relations are allowed. If she were speeding up unevenly they would not be. Steady acceleration, so $v = u + a t$ applies over the whole interval.

4. Do the Working: Rearranging $v = u + a t$ for the time gives $t = (v - u) / a$. That relation has exactly the three things we know and the one we want. $t = (v - u) / a = (9 - 0) / 1.5 = 6 \text{ s}$

(shortened to fit this page)

Checking a rule fits the situation — watch it solved

Watch one worked example (continued)

6. Check Your Answer: A speed divided by an acceleration leaves seconds, so the units are right. Units come out as seconds, and the time is positive.

Answer: She reaches 9 m/s after 6 s.

One thing to check: Divide a speed by an acceleration and what is left is seconds. If your answer came out in metres you picked a relation with a distance in it.

Check it a different way: Work out how far she travels, twice, by two different routes: once from the speeds alone using $(v^2 - u^2) / (2a)$, and once from the time you just found using $ut + \frac{1}{2}at^2$

(shortened to fit this page)

Easy mistake to make

Lots of people fall for this one.

What lots of people think: Applying a single whole-journey relation across a change of acceleration.

Why that feels right: The shortcut looks sufficient because it reproduces the right answer in the cases seen so far, before the decisive physical feature is tested.

Now compare these two cases: A case pair in which the shortcut and the correct model give different answers.

Fix it like this: Predict both cases with the shortcut.

Fix it like this: Work both cases with the correct model.

Fix it like this: Say in one sentence which feature the shortcut ignores.

(shortened to fit this page)

1.7 Checking a rule fits the situation

Checking a rule fits the situation

Check it before you accept it.

Does the answer make sense?

Would this answer make sense in real life?

A check that can never come out wrong is not a check. A real one has to be able to tell you the answer is wrong.

Verify acceleration sign units

Verify constant g assumptions

(shortened to fit this page)

Quick check

Thirty seconds, no notes.

A skateboarder starts from rest on a flat straight path and speeds up steadily at 1.5 m/s^2 . How long does it take her to reach 6 m/s ?

Without looking back, say what your first move would be on the last question.

(shortened to fit this page)

Ready to practise?

You will meet it again here.

Where this idea turns up next: the same idea in a situation that looks nothing like this one. You will meet

(shortened to fit this page)

1.7 Checking a rule fits the situation

Checking a rule fits the situation

Try it with help, then with less, then on your own.

Try it with help

The first move is done for you.

A skateboarder starts from rest on a flat straight path and speeds up steadily at 2 m/s^2 . How long does it take her to reach 10 m/s ?

What you are told: starting velocity = 0 m/s ; acceleration = 2 m/s^2 ; velocity she is aiming for = 10 m/s .

What you want: time taken.

Use the supplied first-move cue and the pre-drawn frame; complete the remaining steps.

The full answer is in the answers appendix — try it first.

Once you have an answer, check it: Verify acceleration sign units; Verify constant g assumptions; Verify dimensions; Verify interval not cumulative; Verify maximum height endpoint; Check you were allowed to

(shortened to fit this page)

Try with less help

One of the cues is gone now.

A skateboarder starts from rest on a flat straight path and speeds up steadily at 2.5 m/s^2 . How long does it take her to reach 12 m/s ?

What you are told: starting velocity = 0 m/s ; acceleration = 2.5 m/s^2 ; velocity she is aiming for = 12 m/s .

What you want: time taken.

One supplied cue has been removed; decide that step yourself and complete the rest.

The full answer is in the answers appendix — try it first.

Once you have an answer, check it: Verify acceleration sign units; Verify constant g assumptions; Verify dimensions; Verify interval not cumulative; Verify maximum height endpoint; Check you were allowed to

(shortened to fit this page)

1.7 Checking a rule fits the situation

Checking a rule fits the situation

Try it with help, then with less, then on your own.

Now try it yourself

No cues this time. You have everything you need.

A skateboarder starts from rest on a flat straight path and speeds up steadily at 3 m/s^2 . How long does it take her to reach 7.5 m/s ?

What you are told: starting velocity = 0 m/s ; acceleration = 3 m/s^2 ; velocity she is aiming for = 7.5 m/s .

What you want: time taken.

No procedural cue is supplied beyond the situation and the required representation.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

WHERE ZERO IS AND WHICH WAY IS POSITIVE



Declared positive direction: right. Re-sign every given quantity against it here.

More ways to see this — Signed axis strip marking the origin, the declared positive direction, and each signed

WHERE IT IS AT EQUAL TIME STEPS

Gaps are equal time steps; unequal gaps mean the speed is changing.



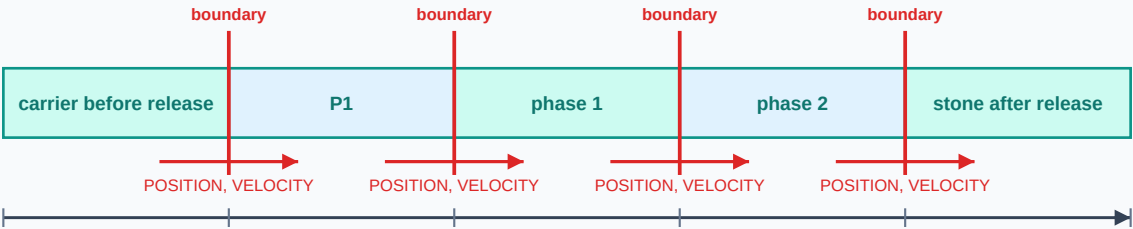
More ways to see this — Strip of successive body positions at equal time intervals along the declared axis. Re

1.7 Checking a rule fits the situation

Checking a rule fits the situation

Different pictures of the same idea.

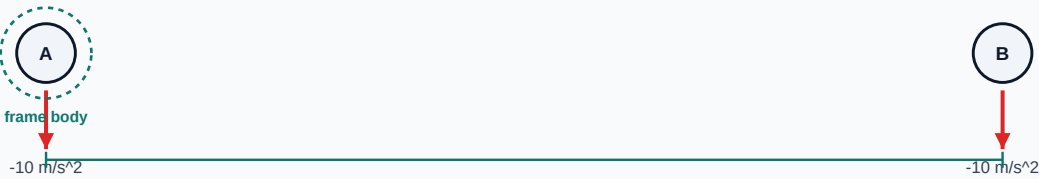
WHAT CARRIES OVER WHEN THE MOTION CHANGES



The end state of one phase is the start state of the next; only the rate may jump.

More ways to draw this — Phase strip with boundary markers and the continuity state labelled across each bound

MEASURED FROM WHICH OBJECT?



More ways to draw this — Relative-frame view marking the frame body and the relative quantity between the two

WHAT YOU KNOW AT THE START AND AT THE END

state variable	stated in the source	to find
acceleration	3 m/s ²	?
displacement	?	?
height	45 m	?
position	?	?
relative acceleration	?	?
time	5 s	?
velocity	0	?

Fill both instants first; the empty cell is what the question asks for.

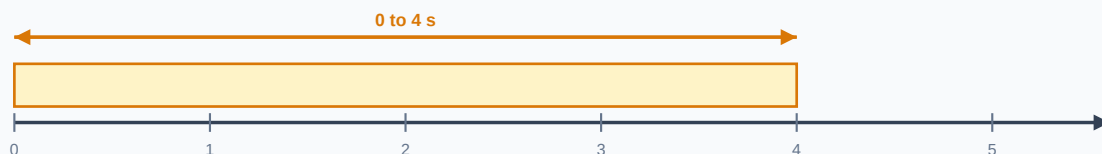
More ways to draw this — Ruled state table with a start row and an end row and the unknown cell marked. Repres

1.7 Checking a rule fits the situation

Checking a rule fits the situation

Different pictures of the same idea.

ONE NAMED SECOND IS A DIFFERENCE



Compute the quantity at each bounding instant, then subtract.

More ways to draw this — Timeline marking the bounding instants of the named interval and the span between the

WHAT EACH LETTER MEANS IN THE PICTURE

DIAGRAM FEATURE



SYMBOL IN THE RELATION

- u** arrow at the start instant
- v** arrow at the end instant
- a** change of the arrow between the instants
- t** span between the two instants

Bind every symbol to a diagram feature before substituting.

More ways to check this — Bridge panel linking each symbol of the relation to the diagram feature it names. Re

WHEN YOU ARE ALLOWED TO USE THIS RULE

model: phy model constant g 1d, phy model moving release, phy model piecewise constant

- | | |
|--|---------------------------|
| ☐ acceleration is constant over the interval | stated / implied / absent |
| ☐ acceleration is constant within each phase | stated / implied / absent |
| ☐ acceleration is constant within each stated phase | stated / implied / absent |
| ☐ acceleration is uniform over the 5 s interval, or the requested quantity is | stated / implied / absent |
| ☐ air resistance neglected | stated / implied / absent |

If any condition is absent the relation below this line may not be used.

More ways to check this — Whether the rule is allowed here gate listing each assumption with its stated, impli

1.7 Checking a rule fits the situation

Checking a rule fits the situation

Different pictures of the same idea.

CHECKS THAT COULD CATCH A MISTAKE

result under test

- ☐ verify acceleration sign units
- ☐ verify constant g assumptions
- ☐ verify dimensions
- ☐ verify interval not cumulative
- ☐ verify maximum height endpoint

Tick only after the check has actually been carried out on the result.

More ways to check this — Verification strip with one tick box per required second, different check. Represent

TWO CASES, ONE THING CHANGED

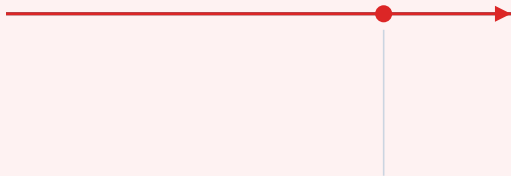
CASE A

the shortcut reproduces the right answer here



CASE B

the shortcut fails here



Decisive feature: A case pair in which the shortcut and the correct model give different answers.

More ways to check this — Two-case contrast panel in which exactly one physical feature differs. Representatio

1.8 MOVING RELEASE

Moving Release

Picture what is happening before you reach for a formula.

Start with a real situation

A real situation before any symbols.

The object does not teleport: position and velocity at the instant of the change belong to both phases.

(shortened to fit this page)

The big idea

The one sentence worth remembering.

Solve one stage at a time, and hand the end of each stage to the start of the next.

At a phase boundary the state is continuous

(shortened to fit this page)

When does this model work?

Check the situation fits before you use the rule.

Name the model, list its assumptions, and mark each assumption as stated, implied or absent.

air resistance neglected

g constant after release

(shortened to fit this page)

WHAT IS ACTUALLY HAPPENING

- 1 Read what physically happens, in order.
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Link the real situation to a drawing before you calculate.

See it step by step

Draw what is happening before you write anything.

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1. Mark every instant at which the motion changes character.
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(shortened to fit this page)

1.8 MOVING RELEASE

Moving Release

One question, solved all the way to a number.

Turn the Picture Into Symbols

Each mark in the drawing gets a name.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

WHICH WAY EACH ARROW POINTS



The same situation drawn out. No new numbers are introduced here.

Watch one worked example

Every step, with the numbers.

A hot-air balloon is rising steadily at 8 m/s when a small sandbag slips off the side. Take gravity as 10 m/s² and ignore air resistance. How long after it slips off does the sandbag stop rising, and how much higher does it get first?

Set it up: you are following the sandbag, from the moment it leaves the balloon, measuring from the point where it slips off, counting upwards as positive.

What you are told: velocity the sandbag has at the moment it slips off = 8 m/s; acceleration due to gravity = 10 m/s². What you want: time until it stops rising.

1. Set Up Your Directions: Call upwards positive, so the sandbag's starting velocity is positive and gravity is negative. Mixing those up is the usual way this goes wrong. Up is positive. Zero is the point where it slips off.
2. See It: Draw two arrows on the sandbag at the moment of release: a velocity arrow pointing up, an acceleration arrow pointing down. They fight each other. Velocity arrow up, acceleration arrow down — so it is still rising but slowing.
3. Which Rule Fits Here?: This is the decisive bit. The sandbag does not start from rest: it leaves the balloon already moving up at the balloon's speed. From then on only gravity acts on it. It keeps the balloon's upward velocity at release, then slows at g until it momentarily stops.

(shortened to fit this page)

MOVING RELEASE — watch it solved

Watch one worked example (continued)

5. Do the Working: From $v^2 = u^2 - 2gh$ with the final velocity zero, the extra rise is $u^2 / 2g$. $\text{rise} = u^2 / (2g) = 8^2 / (2 \cdot 10) = 3.2 \text{ m}$

6. What the Answer Means: It climbs a little further before it starts to come down, which is why a dropped object from a rising balloon does not head straight for the ground. It keeps rising for a moment, then turns round.

Answer: The sandbag keeps rising for another 0.8 s, climbing a further 3.2 m before it starts to fall.

One thing to check: At the very top the sandbag's speed is zero, but its acceleration is still 10 m/s^2 downwards. Zero speed does not mean zero acceleration — if it did, it would hang there forever.

Check it a different way: Check the velocity at the instant you found. Starting at 8 m/s upwards and losing
(shortened to fit this page)

Easy mistake to make

Lots of people fall for this one.

What lots of people think: Applying a single whole-journey relation across a change of acceleration.

Why that feels right: The shortcut looks sufficient because it reproduces the right answer in the cases seen so far, before the decisive physical feature is tested.

Now compare these two cases: A case pair in which the shortcut and the correct model give different answers.

Fix it like this: Predict both cases with the shortcut.

Fix it like this: Work both cases with the correct model.

Fix it like this: Say in one sentence which feature the shortcut ignores.

(shortened to fit this page)

1.8 MOVING RELEASE

Moving Release

Check it before you accept it.

Does the answer make sense?

Would this answer make sense in real life?

A check that can never come out wrong is not a check. A real one has to be able to tell you the answer is wrong.

Check you were allowed to use that rule

Verify release state continuity

(shortened to fit this page)

Quick check

Thirty seconds, no notes.

A hot-air balloon is rising steadily at 10 m/s when a small sandbag slips off the side. Take gravity as 10 m/s^2 and ignore air resistance. How long after it slips off does the sandbag stop rising, and how much higher does it get first?

Without looking back, say what your first move would be on the last question.

(shortened to fit this page)

Ready to practise?

You will meet it again here.

Where this idea turns up next: the same idea in a situation that looks nothing like this one. You will meet

(shortened to fit this page)

1.8 MOVING RELEASE

Moving Release

Try it with help, then with less, then on your own.

Try it with help

The first move is done for you.

A hot-air balloon is rising steadily at 5 m/s when a small sandbag slips off the side. Take gravity as 10 m/s^2 and ignore air resistance. How long after it slips off does the sandbag stop rising, and how much higher does it get first?

What you are told: velocity the sandbag has at the moment it slips off = 5 m/s; acceleration due to gravity = 10 m/s^2 . What you want: time until it stops rising.

Use the supplied first-move cue and the pre-drawn frame; complete the remaining steps.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

Try with less help

One of the cues is gone now.

A hot-air balloon is rising steadily at 12 m/s when a small sandbag slips off the side. Take gravity as 10 m/s^2 and ignore air resistance. How long after it slips off does the sandbag stop rising, and how much higher does it get first?

What you are told: velocity the sandbag has at the moment it slips off = 12 m/s; acceleration due to gravity = 10 m/s^2 . What you want: time until it stops rising.

One supplied cue has been removed; decide that step yourself and complete the rest.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

1.8 MOVING RELEASE

Moving Release

Try it with help, then with less, then on your own.

Now try it yourself

No cues this time. You have everything you need.

A hot-air balloon is rising steadily at 6 m/s when a small sandbag slips off the side. Take gravity as 10 m/s² and ignore air resistance. How long after it slips off does the sandbag stop rising, and how much higher does it get first?

What you are told: velocity the sandbag has at the moment it slips off = 6 m/s; acceleration due to gravity = 10 m/s². What you want: time until it stops rising.

No procedural cue is supplied beyond the situation and the required representation.

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(shortened to fit this page)

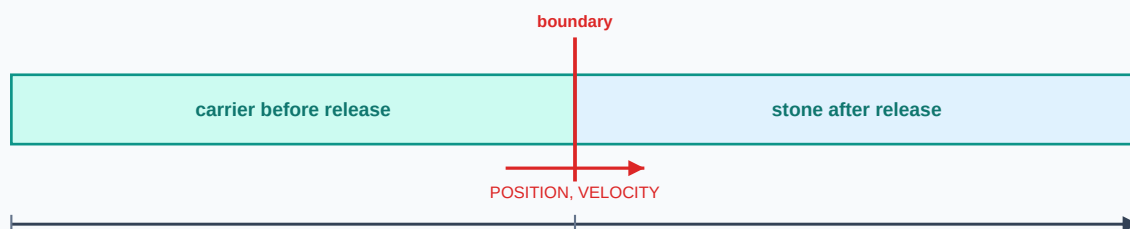
WHERE IT IS AT EQUAL TIME STEPS

Gaps are equal time steps; unequal gaps mean the speed is changing.



More ways to see this — Strip of successive body positions at equal time intervals along the declared axis. Re

WHAT CARRIES OVER WHEN THE MOTION CHANGES



The end state of one phase is the start state of the next; only the rate may jump.

More ways to draw this — Phase strip with boundary markers and the continuity state labelled across each bound

1.8 MOVING RELEASE

Moving Release

Different pictures of the same idea.

WHAT YOU KNOW AT THE START AND AT THE END

state variable	stated in the source	to find
acceleration	?	?
position	?	?
time	?	?
velocity	8 m/s	?

Fill both instants first; the empty cell is what the question asks for.

More ways to draw this — Ruled state table with a start row and an end row and the unknown cell marked. Repres

WHEN YOU ARE ALLOWED TO USE THIS RULE

model: phy model constant g 1d, phy model moving release

- ☐ air resistance neglected
- ☐ g constant after release
- ☐ released body inherits carrier velocity at release
- stated / implied / absent
- stated / implied / absent
- stated / implied / absent

If any condition is absent the relation below this line may not be used.

More ways to check this — Whether the rule is allowed here gate listing each assumption with its stated, impli

CHECKS THAT COULD CATCH A MISTAKE

result under test

- ☐ check you were allowed to use that rule
- ☐ verify release state continuity
- ☐ verify source representation present

Tick only after the check has actually been carried out on the result.

More ways to check this — Verification strip with one tick box per required second, different check. Represent

1.8 MOVING RELEASE

Moving Release

Different pictures of the same idea.

TWO CASES, ONE THING CHANGED

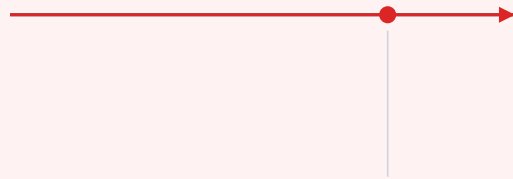
CASE A

the shortcut reproduces the right answer here



CASE B

the shortcut fails here



Decisive feature: A case pair in which the shortcut and the correct model give different answers.

More ways to check this — Two-case contrast panel in which exactly one physical feature differs. Representatio

1.9 MULTIPHASE KINEMATICS

Multiphase Kinematics

Picture what is happening before you reach for a formula.

Start with a real situation

A real situation before any symbols.

The object does not teleport: position and velocity at the instant of the change belong to both phases.

(shortened to fit this page)

The big idea

The one sentence worth remembering.

Solve one stage at a time, and hand the end of each stage to the start of the next.

At a phase boundary the state is continuous

(shortened to fit this page)

When does this model work?

Check the situation fits before you use the rule.

Name the model, list its assumptions, and mark each assumption as stated, implied or absent.

acceleration is constant within each phase

acceleration is constant within each stated phase

(shortened to fit this page)

WHAT IS ACTUALLY HAPPENING

- 1 Read what physically happens, in order.
- 2 Mark the start state and the end state.
- 3 Mark the quantity the question asks for.



Link the real situation to a drawing before you calculate.

See it step by step

Draw what is happening before you write anything.

State the origin, the positive direction, and which object the frame is attached to.

Draw a phase strip and mark the quantity that must carry over unchanged.

1. Mark every instant at which the motion changes character.
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(shortened to fit this page)

1.9 MULTIPHASE KINEMATICS

Multiphase Kinematics

One question, solved all the way to a number.

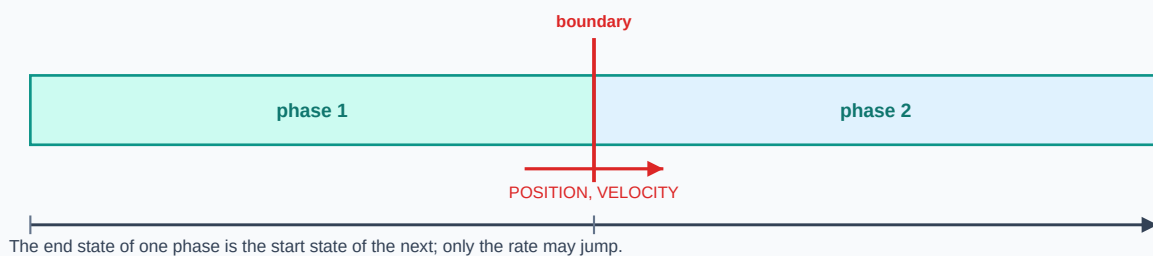
Turn the Picture Into Symbols

Each mark in the drawing gets a name.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

WHAT CARRIES OVER WHEN THE MOTION CHANGES



The same situation drawn out. No new numbers are introduced here.

Watch one worked example

Every step, with the numbers.

A cart is already rolling at 5 m/s along a straight track. For the first 3 s it speeds up steadily at 2 m/s^2 . Then the track changes and for the next 2 s it slows steadily at -1 m/s^2 . How fast is the cart moving at the very end?

Set it up: you are following the cart, measuring from where the timing starts, counting the way the cart is already rolling as positive.

What you are told: velocity at the start = 5 m/s; acceleration in the first part = 2 m/s^2 ; how long the first part lasts = 3 s; acceleration in the second part = -1 m/s^2 ; how long the second part lasts = 2 s. What you want: velocity at the end.

1. Set Up Your Directions: One direction for the whole problem: the way the cart is already rolling is positive. Speeding up is then positive and slowing down is negative. Forwards is positive, for both parts.

2. See It: Draw the journey as two blocks with a line between them. The moment the track changes is the boundary, and it belongs to both parts. Part 1, boundary, part 2. The velocity at the boundary is shared.

3. Which Rule Fits Here?: The acceleration is steady inside each part but jumps between them, so you cannot use one relation across the whole journey. You have to do the parts one at a time. Steady acceleration within each part, handled separately, joined at the boundary.

(shortened to fit this page)

MULTIPHASE KINEMATICS — watch it solved

Watch one worked example (continued)

5. Do the Working: Part 2 starts at exactly the velocity part 1 finished with — not at zero. That carried-over velocity is the whole point of a two-part problem. $v_2 = v_1 + a_2 \cdot t_2 = 11 + -1 \cdot 2 = 9 \text{ m/s}$

6. What the Answer Means: Still positive, so the cart is still rolling forwards at the end — just not as fast as at the boundary. Faster than it started, slower than at the boundary, still going forwards.

Answer: The cart reaches 11 m/s at the boundary and finishes at 9 m/s, still rolling forwards.

One thing to check: Part 2 has to start at the velocity part 1 ended with. If you started part 2 from zero you threw the boundary state away, and the answer will come out far too small.

Check it a different way: Check the first part a different way. Work out how far the cart travels in part 1

(shortened to fit this page)

Easy mistake to make

Lots of people fall for this one.

What lots of people think: Applying a single whole-journey relation across a change of acceleration.

Why that feels right: The shortcut looks sufficient because it reproduces the right answer in the cases seen so far, before the decisive physical feature is tested.

Now compare these two cases: A case pair in which the shortcut and the correct model give different answers.

Fix it like this: Predict both cases with the shortcut.

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1.9 MULTIPHASE KINEMATICS

Multiphase Kinematics

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Does the answer make sense?

Would this answer make sense in real life?

A check that can never come out wrong is not a check. A real one has to be able to tell you the answer is wrong.

Check you were allowed to use that rule

Verify phase state continuity

(shortened to fit this page)

Quick check

Thirty seconds, no notes.

A cart is already rolling at 8 m/s along a straight track. For the first 6 s it speeds up steadily at 2 m/s^2 . Then the track changes and for the next 2 s it slows steadily at -1 m/s^2 . How fast is the cart moving at the very end?

Without looking back, say what your first move would be on the last question.

(shortened to fit this page)

Ready to practise?

You will meet it again here.

Where this idea turns up next: the same idea in a situation that looks nothing like this one. You will meet

(shortened to fit this page)

1.9 MULTIPHASE KINEMATICS

Multiphase Kinematics

Try it with help, then with less, then on your own.

Try it with help

The first move is done for you.

A cart is already rolling at 4 m/s along a straight track. For the first 4 s it speeds up steadily at 1.5 m/s^2 . Then the track changes and for the next 3 s it slows steadily at -2 m/s^2 . How fast is the cart moving at the very end?

What you are told: velocity at the start = 4 m/s; acceleration in the first part = 1.5 m/s^2 ; how long the first part lasts = 4 s; acceleration in the second part = -2 m/s^2 ; how long the second part lasts = 3 s. What you want: velocity at the end.

Use the supplied first-move cue and the pre-drawn frame; complete the remaining steps.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

Try with less help

One of the cues is gone now.

A cart is already rolling at 6 m/s along a straight track. For the first 2 s it speeds up steadily at 3 m/s^2 . Then the track changes and for the next 4 s it slows steadily at -1.5 m/s^2 . How fast is the cart moving at the very end?

What you are told: velocity at the start = 6 m/s; acceleration in the first part = 3 m/s^2 ; how long the first part lasts = 2 s; acceleration in the second part = -1.5 m/s^2 ; how long the second part lasts = 4 s. What you want: velocity at the end.

One supplied cue has been removed; decide that step yourself and complete the rest.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

1.9 MULTIPHASE KINEMATICS

Try it with help, then with less, then on your own.

Now try it yourself

No cues this time. You have everything you need.

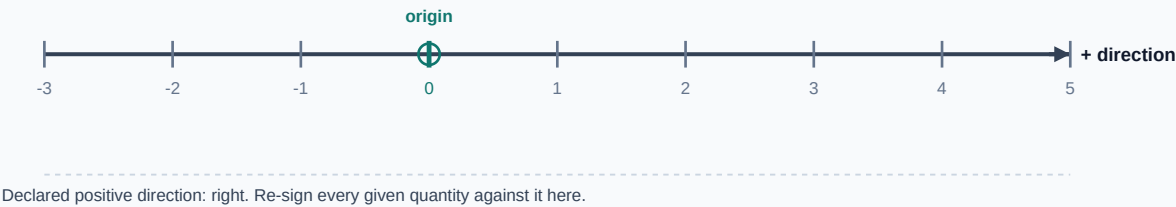
A cart is already rolling at 2 m/s along a straight track. For the first 5 s it speeds up steadily at 2.5 m/s². Then the track changes and for the next 1 s it slows steadily at -0.5 m/s². How fast is the cart moving at the very end?

What you are told: velocity at the start = 2 m/s; acceleration in the first part = 2.5 m/s²; how long the first part lasts = 5 s; acceleration in the second part = -0.5 m/s²; how long the second part lasts = 1 s. What you want: velocity at the end.

No procedural cue is supplied beyond the situation and the required representation.

(shortened to fit this page)

WHERE ZERO IS AND WHICH WAY IS POSITIVE



More ways to see this — Signed axis strip marking the origin, the declared positive direction, and each signed

WHERE IT IS AT EQUAL TIME STEPS

Gaps are equal time steps; unequal gaps mean the speed is changing.



More ways to see this — Strip of successive body positions at equal time intervals along the declared axis. Re

1.9 MULTIPHASE KINEMATICS

Different pictures of the same idea.

WHAT YOU KNOW AT THE START AND AT THE END

state variable	stated in the source	to find
acceleration	?	?
displacement	?	?
time	?	?
velocity	?	?

Fill both instants first; the empty cell is what the question asks for.

More ways to draw this — Ruled state table with a start row and an end row and the unknown cell marked. Repres

WHEN YOU ARE ALLOWED TO USE THIS RULE

model: phy model piecewise constant acceleration		
☐ acceleration is constant within each phase		stated / implied / absent
☐ acceleration is constant within each stated phase		stated / implied / absent
☐ phase-boundary state is propagated continuously		stated / implied / absent
☐ velocity and position are continuous at the phase boundary		stated / implied / absent

If any condition is absent the relation below this line may not be used.

More ways to check this — Whether the rule is allowed here gate listing each assumption with its stated, impli

CHECKS THAT COULD CATCH A MISTAKE

result under test	
☐ check you were allowed to use that rule	
☐ verify phase state continuity	
☐ verify source representation present	

Tick only after the check has actually been carried out on the result.

More ways to check this — Verification strip with one tick box per required second, different check. Represent

1.9 MULTIPHASE KINEMATICS

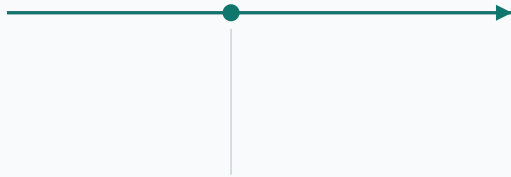
Multiphase Kinematics

Different pictures of the same idea.

TWO CASES, ONE THING CHANGED

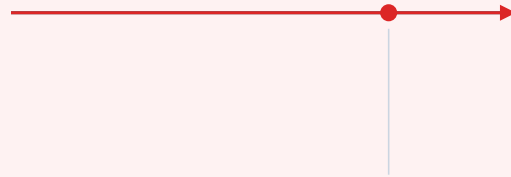
CASE A

the shortcut reproduces the right answer here



CASE B

the shortcut fails here



Decisive feature: A case pair in which the shortcut and the correct model give different answers.

More ways to check this — Two-case contrast panel in which exactly one physical feature differs. Representatio

1.10 NTH SECOND INTERVAL

Nth Second Interval

Picture what is happening before you reach for a formula.

Start with a real situation

A real situation before any symbols.

Ask what is being assumed constant, and whether the situation says it is.

(shortened to fit this page)

The big idea

The one sentence worth remembering.

Before using a relation, say in words what it assumes and point to where the question says that is true.

A relation is only valid while its assumptions

(shortened to fit this page)

When does this model work?

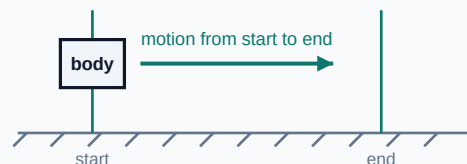
Check the situation fits before you use the rule.

Name the model, list its assumptions, and mark each assumption as stated, implied or absent.

(shortened to fit this page)

WHAT IS ACTUALLY HAPPENING

- 1 Read what physically happens, in order.
- 2 Mark the start state and the end state.
- 3 Mark the quantity the question asks for.



Link the real situation to a drawing before you calculate.

See it step by step

Draw what is happening before you write anything.

State the origin, the positive direction, and which object the frame is attached to.

Check each assumption of the candidate model against a stated condition.

1. List the candidate physical models for this situation.

2. Write each model's assumptions as a checklist.

(shortened to fit this page)

1.10 NTH SECOND INTERVAL

Nth Second Interval

One question, solved all the way to a number.

Turn the Picture Into Symbols

Each mark in the drawing gets a name.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

WHAT YOU KNOW AT THE START AND AT THE END

state variable	stated in the source	to find
acceleration	3 m/s^2	?
displacement	?	?

Fill both instants first; the empty cell is what the question asks for.

The same situation drawn out. No new numbers are introduced here.

Watch one worked example

Every step, with the numbers.

A go-kart is moving at 2 m/s and then speeds up steadily at 3 m/s^2. How far does it travel during the 5th second alone — that is, between the end of second 5 minus one and the end of second 5?

Set it up: you are following the go-kart, measuring from where it is when the clock starts, counting the way it is moving as positive.

What you are told: velocity when the clock starts = 2 m/s; acceleration = 3 m/s^2; which second we want = 5. What you want: distance travelled during that one second.

1. Set Up Your Directions: One direction, forwards, and one clock started at the moment we are given the velocity. Forwards is positive and the clock starts at zero.

2. See It: Draw the clock as a line and shade just the one-second slot we want. It sits between two instants, it is not the whole journey. Shade the slot from the end of one second to the end of the next.

3. Which Rule Fits Here?: 'During the nth second' means between $t = n - 1$ and $t = n$. That is a difference between two totals, not a total. A named second is an interval: distance up to n , minus distance up to $n - 1$.

4. Do the Working: Subtracting the two totals tidies up to $u + a(2n - 1)/2$. Every term is a distance, because the interval is exactly one second long. $s_n = u + a(2n - 1)/2 = 2 + 3(2 \cdot 5 - 1)/2 = 15.5 \text{ m}$

(shortened to fit this page)

NTH SECOND INTERVAL — watch it solved

Watch one worked example (continued)

6. Check Your Answer: It should be bigger than the distance in the first second and smaller than the whole journey so far. Bigger than the earlier seconds, smaller than the total.

Answer: During that one second alone the go-kart covers 15.5 m.

One thing to check: This answer is a distance in metres, not a velocity — even though the working looks like a velocity calculation. It looks that way because the interval is exactly one second long.

Check it a different way: Do it the long way instead. Work out the total distance up to the end of second 5,
(shortened to fit this page)

Easy mistake to make

Lots of people fall for this one.

What lots of people think: Using a constant-rate relation for a situation that changes rate part-way through.

Why that feels right: The shortcut looks sufficient because it reproduces the right answer in the cases seen so far, before the decisive physical feature is tested.

Now compare these two cases: A case pair in which the shortcut and the correct model give different answers.

Fix it like this: Predict both cases with the shortcut.

Fix it like this: Work both cases with the correct model.

Fix it like this: Say in one sentence which feature the shortcut ignores.

(shortened to fit this page)

1.10 NTH SECOND INTERVAL

Nth Second Interval

Check it before you accept it.

Does the answer make sense?

Would this answer make sense in real life?

A check that can never come out wrong is not a check. A real one has to be able to tell you the answer is wrong.

Verify dimensions

Verify interval not cumulative

(shortened to fit this page)

Quick check

Thirty seconds, no notes.

A go-kart is moving at 4 m/s and then speeds up steadily at 3 m/s². How far does it travel during the 5th second alone — that is, between the end of second 5 minus one and the end of second 5?

Without looking back, say what your first move would be on the last question.

(shortened to fit this page)

Ready to practise?

You will meet it again here.

Where this idea turns up next: the same idea in a situation that looks nothing like this one. You will meet

(shortened to fit this page)

Try it with help

The first move is done for you.

A go-kart is moving at 0 m/s and then speeds up steadily at 2 m/s². How far does it travel during the 4th second alone — that is, between the end of second 4 minus one and the end of second 4?

What you are told: velocity when the clock starts = 0 m/s; acceleration = 2 m/s²; which second we want = 4. What you want: distance travelled during that one second.

Use the supplied first-move cue and the pre-drawn frame; complete the remaining steps.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

1.10 NTH SECOND INTERVAL

Nth Second Interval

Try it with help, then with less, then on your own.

Try with less help

One of the cues is gone now.

A go-kart is moving at 3 m/s and then speeds up steadily at 4 m/s². How far does it travel during the 6th second alone — that is, between the end of second 6 minus one and the end of second 6?

What you are told: velocity when the clock starts = 3 m/s; acceleration = 4 m/s²; which second we want = 6. What you want: distance travelled during that one second.

One supplied cue has been removed; decide that step yourself and complete the rest.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

Now try it yourself

No cues this time. You have everything you need.

A go-kart is moving at 1 m/s and then speeds up steadily at 1.5 m/s². How far does it travel during the 3th second alone — that is, between the end of second 3 minus one and the end of second 3?

What you are told: velocity when the clock starts = 1 m/s; acceleration = 1.5 m/s²; which second we want = 3. What you want: distance travelled during that one second.

No procedural cue is supplied beyond the situation and the required representation.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

1.10 NTH SECOND INTERVAL

Nth Second Interval

Different pictures of the same idea.

ONE NAMED SECOND IS A DIFFERENCE



Compute the quantity at each bounding instant, then subtract.

More ways to draw this — Timeline marking the bounding instants of the named interval and the span between the

WHAT EACH LETTER MEANS IN THE PICTURE

DIAGRAM FEATURE



SYMBOL IN THE RELATION

- u** arrow at the start instant
- a** change of the arrow between the instants

Bind every symbol to a diagram feature before substituting.

More ways to check this — Bridge panel linking each symbol of the relation to the diagram feature it names. Re

WHEN YOU ARE ALLOWED TO USE THIS RULE

model: phy model uniform acceleration

☐ the stated acceleration is constant over the interval stated / implied / absent

If any condition is absent the relation below this line may not be used.

More ways to check this — Whether the rule is allowed here gate listing each assumption with its stated, impli

1.10 NTH SECOND INTERVAL

Nth Second Interval

Different pictures of the same idea.

CHECKS THAT COULD CATCH A MISTAKE

result under test

- ☐ verify dimensions
- ☐ verify interval not cumulative
- ☐ check you were allowed to use that rule

Tick only after the check has actually been carried out on the result.

More ways to check this — Verification strip with one tick box per required second, different check. Represent

TWO CASES, ONE THING CHANGED

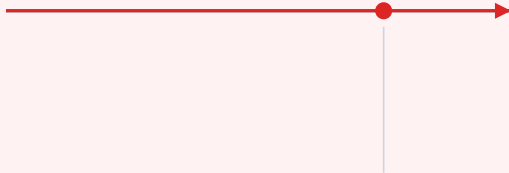
CASE A

the shortcut reproduces the right answer here



CASE B

the shortcut fails here



Decisive feature: A case pair in which the shortcut and the correct model give different answers.

More ways to check this — Two-case contrast panel in which exactly one physical feature differs. Representatio

1.11 RELATIVE ACCELERATION SAME G

Relative Acceleration Same G

Picture what is happening before you reach for a formula.

Start with a real situation

A real situation before any symbols.

Ask what is being assumed constant, and whether the situation says it is.

(shortened to fit this page)

The big idea

The one sentence worth remembering.

Before using a relation, say in words what it assumes and point to where the question says that is true.

A relation is only valid while its assumptions

(shortened to fit this page)

When does this model work?

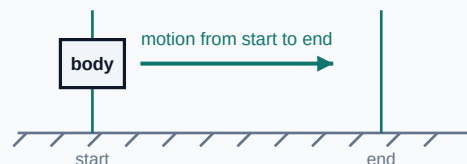
Check the situation fits before you use the rule.

Name the model, list its assumptions, and mark each assumption as stated, implied or absent.

(shortened to fit this page)

WHAT IS ACTUALLY HAPPENING

- 1 Read what physically happens, in order.
- 2 Mark the start state and the end state.
- 3 Mark the quantity the question asks for.



Link the real situation to a drawing before you calculate.

See it step by step

Draw what is happening before you write anything.

State the origin, the positive direction, and which object the frame is attached to.

Check each assumption of the candidate model against a stated condition.

1. List the candidate physical models for this situation.

2. Write each model's assumptions as a checklist.

(shortened to fit this page)

1.11 RELATIVE ACCELERATION SAME G

Relative Acceleration Same G

One question, solved all the way to a number.

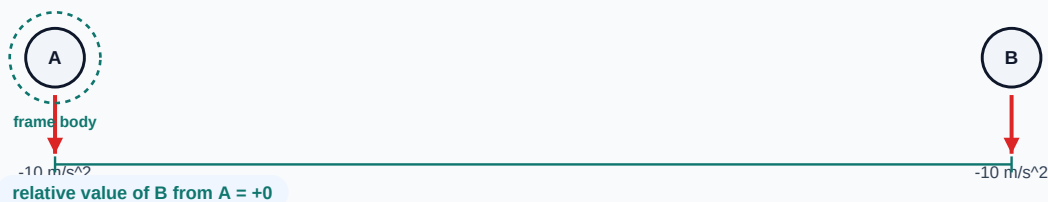
Turn the Picture Into Symbols

Each mark in the drawing gets a name.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

MEASURED FROM WHICH OBJECT?



The same situation drawn out. No new numbers are introduced here.

Watch one worked example

Every step, with the numbers.

Two stones are released from rest at the same instant, one from 40 m up and the other from 20 m up. Take gravity as 10 m/s^2 and ignore air resistance. What is the acceleration of the second stone as seen by someone falling with the first one?

Set it up: you are following both stones, watched from the first one, measuring from the release point of the first stone, counting upwards as positive.

What you are told: acceleration due to gravity = 10 m/s^2 ; height the first stone starts from = 40 m; height the second stone starts from = 20 m; starting velocity of the first stone = 0 m/s; starting velocity of the second stone = 0 m/s; any moment during the fall, used only for checking = 2 s. What you want: acceleration of the second stone relative to the first.

1. Set Up Your Directions: This question does not ask what the stones do from the ground — it asks what one does as seen from the other. Naming who is watching is the whole job here. Up is positive, and the observer is falling with the first stone.

2. See It: Draw an acceleration arrow on each stone. Both arrows point down and both are the same length. Two identical downward arrows — one on each stone.

3. Which Rule Fits Here?: With air resistance ignored, gravity gives every object the same acceleration whatever its mass and whatever height it started from. The starting heights change where they are, not

(shortened to fit this page)

RELATIVE ACCELERATION SAME G — watch it solved

Watch one worked example (continued)

4. Do the Working: Up is positive and gravity pulls down, so the first stone's acceleration is negative g .
 $a_A = -g = -10 = -10 \text{ m/s}^2$

5. Do the Working: The second stone gets exactly the same treatment from gravity, so its acceleration is the same negative g . $a_B = 0 - g = 0 - 10 = -10 \text{ m/s}^2$

6. Do the Working: 'As seen from the first stone' means subtract the first stone's acceleration from the second stone's. $a_{\text{rel}} = a_B - a_A = -10 - (-10) = 0 \text{ m/s}^2$

7. What the Answer Means: Zero relative acceleration means the gap between the two stones changes at a steady rate — here it does not change at all, since both start from rest. Neither stone gains on the other, so from one of them the other appears to hang still.

Answer: The relative acceleration is 0 m/s^2 : from the first stone, the second one does not appear to accelerate at all.

One thing to check: A zero answer here does not mean nothing is accelerating. Each stone really is speeding up at 10 m/s^2 downwards — they just do it identically, so neither gains on the other.

Check it a different way: Compare their velocities at any moment, say 2 s after release. Each one is its starting velocity minus g times the time. Subtract them and the g -times-time parts cancel, so the gap

(shortened to fit this page)

1.11 RELATIVE ACCELERATION SAME G

Relative Acceleration Same G

The shortcut that looks easier, and why it breaks.

Easy mistake to make

Lots of people fall for this one.

What lots of people think: Using a constant-rate relation for a situation that changes rate part-way through.

Why that feels right: The shortcut looks sufficient because it reproduces the right answer in the cases seen so far, before the decisive physical feature is tested.

Now compare these two cases: A case pair in which the shortcut and the correct model give different answers.

Fix it like this: Predict both cases with the shortcut.

Fix it like this: Work both cases with the correct model.

Fix it like this: Say in one sentence which feature the shortcut ignores.

(shortened to fit this page)

Does the answer make sense?

Would this answer make sense in real life?

A check that can never come out wrong is not a check. A real one has to be able to tell you the answer is wrong.

Check you were allowed to use that rule

(shortened to fit this page)

Quick check

Thirty seconds, no notes.

Two stones are released from rest at the same instant, one from 60 m up and the other from 20 m up. Take gravity as 10 m/s^2 and ignore air resistance. What is the acceleration of the second stone as seen by someone falling with the first one?

Without looking back, say what your first move would be on the last question.

(shortened to fit this page)

1.11 RELATIVE ACCELERATION SAME G

Relative Acceleration Same G

Where this idea turns up next.

Ready to practise?

You will meet it again here.

Where this idea turns up next: the same idea in a situation that looks nothing like this one. You will meet

(shortened to fit this page)

Try it with help

The first move is done for you.

Two stones are released from rest at the same instant, one from 60 m up and the other from 15 m up. Take gravity as 9.8 m/s^2 and ignore air resistance. What is the acceleration of the second stone as seen by someone falling with the first one?

What you are told: acceleration due to gravity = 9.8 m/s^2 ; height the first stone starts from = 60 m; height the second stone starts from = 15 m; starting velocity of the first stone = 0 m/s; starting velocity of the second stone = 0 m/s; any moment during the fall, used only for checking = 3 s. What you want: acceleration of the second stone relative to the first.

Use the supplied first-move cue and the pre-drawn frame; complete the remaining steps.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

1.11 RELATIVE ACCELERATION SAME G

Relative Acceleration Same G

Try it with help, then with less, then on your own.

Try with less help

One of the cues is gone now.

Two stones are released from rest at the same instant, one from 80 m up and the other from 35 m up. Take gravity as 10 m/s^2 and ignore air resistance. What is the acceleration of the second stone as seen by someone falling with the first one?

What you are told: acceleration due to gravity = 10 m/s^2 ; height the first stone starts from = 80 m; height the second stone starts from = 35 m; starting velocity of the first stone = 0 m/s ; starting velocity of the second stone = 0 m/s ; any moment during the fall, used only for checking = 2 s. What you want: acceleration of the second stone relative to the first.

One supplied cue has been removed; decide that step yourself and complete the rest.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

Now try it yourself

No cues this time. You have everything you need.

Two stones are released from rest at the same instant, one from 40 m up and the other from 50 m up. Take gravity as 9.8 m/s^2 and ignore air resistance. What is the acceleration of the second stone as seen by someone falling with the first one?

What you are told: acceleration due to gravity = 9.8 m/s^2 ; height the first stone starts from = 40 m; height the second stone starts from = 50 m; starting velocity of the first stone = 0 m/s ; starting velocity of the second stone = 0 m/s ; any moment during the fall, used only for checking = 3 s. What you want: acceleration of the second stone relative to the first.

No procedural cue is supplied beyond the situation and the required representation.

(shortened to fit this page)

1.11 RELATIVE ACCELERATION SAME G

Relative Acceleration Same G

Different pictures of the same idea.

WHEN YOU ARE ALLOWED TO USE THIS RULE

model: phy model relative motion same g

☐ same uniform gravitational field stated / implied / absent

If any condition is absent the relation below this line may not be used.

More ways to check this — Whether the rule is allowed here gate listing each assumption with its stated, impli

CHECKS THAT COULD CATCH A MISTAKE

result under test

- ☐ ▶ check you were allowed to use that rule
- ☐ ▶ verify relative acceleration cancellation

Tick only after the check has actually been carried out on the result.

More ways to check this — Verification strip with one tick box per required second, different check. Represent

TWO CASES, ONE THING CHANGED

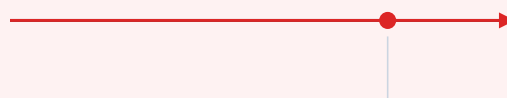
CASE A

the shortcut reproduces the right answer her
e



CASE B

the shortcut fails here



Decisive feature: A case pair in which the shortcut and the correct model give different answers.

More ways to check this — Two-case contrast panel in which exactly one physical feature differs. Representatio

1.12 REPRESENTATION TRANSLATE

Representation Translate

Picture what is happening before you reach for a formula.

Start with a real situation

A real situation before any symbols.

A graph is a record of the motion, not a picture of the path.

(shortened to fit this page)

The big idea

The one sentence worth remembering.

Ask first: do I need how fast this is changing, or how much it has changed in total?

On a graph of a state variable against time, slope is a rate and signed area is a change; which one

(shortened to fit this page)

When does this model work?

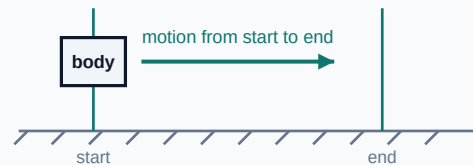
Check the situation fits before you use the rule.

Name the model, list its assumptions, and mark each assumption as stated, implied or absent.

(shortened to fit this page)

WHAT IS ACTUALLY HAPPENING

- 1 Read what physically happens, in order.
- 2 Mark the start state and the end state.
- 3 Mark the quantity the question asks for.



Link the real situation to a drawing before you calculate.

See it step by step

Draw what is happening before you write anything.

State the origin, the positive direction, and which object the frame is attached to.

Decode slope and area separately and say what each one is physically.

1. Read and state both axis labels and units.
2. Decide whether the question asks for a rate or an accumulated change.

(shortened to fit this page)

1.12 REPRESENTATION TRANSLATE

Representation Translate

One question, solved all the way to a number.

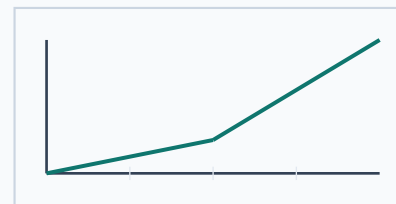
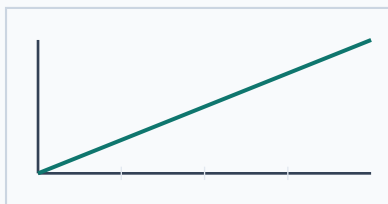
Turn the Picture Into Symbols

Each mark in the drawing gets a name.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

THE CHOICES, DRAWN ON THE SAME AXES



Reject each option by naming the feature it gets wrong.

The same situation drawn out. No new numbers are introduced here.

Watch one worked example

Every step, with the numbers.

A velocity-time graph for a trolley is a straight line. At 0 s the line reads 2 m/s, and at 4 s it reads 14 m/s. What is the trolley's acceleration?

Set it up: you are following the trolley, measuring from the start of the timing, counting the direction the graph calls positive velocity as positive.

What you are told: velocity at the first instant = 2 m/s; first instant = 0 s; velocity at the second instant = 14 m/s; second instant = 4 s. What you want: acceleration.

1. Set Up Your Directions: Read both axis labels first. The up axis is velocity, the across axis is time — so the steepness has velocity on top and time underneath. Up axis: velocity in m/s. Across axis: time in s.
2. See It: Mark the two points on the line and draw the step between them: across for the time, up for the velocity gained. A right-angled step under the line: across is the time taken, up is the velocity gained.
3. Which Rule Fits Here?: The line is straight, so the velocity changes at the same rate the whole way. That is exactly what constant acceleration means. Straight line on a velocity-time graph means steady acceleration, and the steepness is that acceleration.
4. Do the Working: Steepness means change in the up quantity divided by change in the across quantity.

$$a = (v_2 - v_1) / (t_2 - t_1) = (14 - 2) / (4 - 0) = 3 \text{ m/s}^2$$

(shortened to fit this page)

REPRESENTATION TRANSLATE — watch it solved

Watch one worked example (continued)

direction it is already going. It gains that much velocity every second.

6. Check Your Answer: Metres per second, divided by seconds, is metres per second squared — the units of an acceleration. The units come out as m/s^2 .

Answer: The trolley's acceleration is 3 m/s^2 — it gains that much velocity every second.

One thing to check: The steepness of a velocity-time line is an acceleration, in m/s^2 . If your answer came out in m/s you read the height of the line instead of how steeply it climbs.

Check it a different way: Run it backwards through the other graph. On an acceleration-time graph, the
(shortened to fit this page)

Easy mistake to make

Lots of people fall for this one.

What lots of people think: Reading a velocity-time graph as if it were a picture of the path travelled.

Why that feels right: The shortcut looks sufficient because it reproduces the right answer in the cases seen so far, before the decisive physical feature is tested.

Now compare these two cases: A case pair in which the shortcut and the correct model give different answers.

Fix it like this: Predict both cases with the shortcut.

Fix it like this: Work both cases with the correct model.

Fix it like this: Say in one sentence which feature the shortcut ignores.

(shortened to fit this page)

1.12 REPRESENTATION TRANSLATE

Representation Translate

Check it before you accept it.

Does the answer make sense?

Would this answer make sense in real life?

A check that can never come out wrong is not a check. A real one has to be able to tell you the answer is wrong.

Verify dimensions

Verify graph slope sign

(shortened to fit this page)

Quick check

Thirty seconds, no notes.

A velocity-time graph for a trolley is a straight line. At 1 s the line reads 3 m/s, and at 4 s it reads 16 m/s. What is the trolley's acceleration?

Without looking back, say what your first move would be on the last question.

(shortened to fit this page)

Ready to practise?

You will meet it again here.

Where this idea turns up next: the same idea in a situation that looks nothing like this one. You will meet

(shortened to fit this page)

1.12 REPRESENTATION TRANSLATE

Representation Translate

Try it with help, then with less, then on your own.

Try it with help

The first move is done for you.

A velocity-time graph for a trolley is a straight line. At 1 s the line reads 0 m/s, and at 5 s it reads 12 m/s. What is the trolley's acceleration?

What you are told: velocity at the first instant = 0 m/s; first instant = 1 s; velocity at the second instant = 12 m/s; second instant = 5 s. What you want: acceleration.

Use the supplied first-move cue and the pre-drawn frame; complete the remaining steps.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

Try with less help

One of the cues is gone now.

A velocity-time graph for a trolley is a straight line. At 2 s the line reads 4 m/s, and at 6 s it reads 10 m/s. What is the trolley's acceleration?

What you are told: velocity at the first instant = 4 m/s; first instant = 2 s; velocity at the second instant = 10 m/s; second instant = 6 s. What you want: acceleration.

One supplied cue has been removed; decide that step yourself and complete the rest.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

1.12 REPRESENTATION TRANSLATE

Representation Translate

Try it with help, then with less, then on your own.

Now try it yourself

No cues this time. You have everything you need.

A velocity-time graph for a trolley is a straight line. At 0 s the line reads 1 m/s, and at 8 s it reads 9 m/s. What is the trolley's acceleration?

What you are told: velocity at the first instant = 1 m/s; first instant = 0 s; velocity at the second instant = 9 m/s; second instant = 8 s. What you want: acceleration.

No procedural cue is supplied beyond the situation and the required representation.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

POSITION AGAINST TIME: STEEPNESS IS SPEED



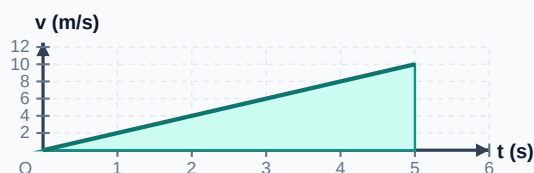
More ways to draw this — Position-time graph with labelled axes and a rise-over-run slope triangle. Representa

STEEPNESS OR AREA? ONE GRAPH, TWO READINGS

READ AS SLOPE = a rate



READ AS SIGNED AREA = a change



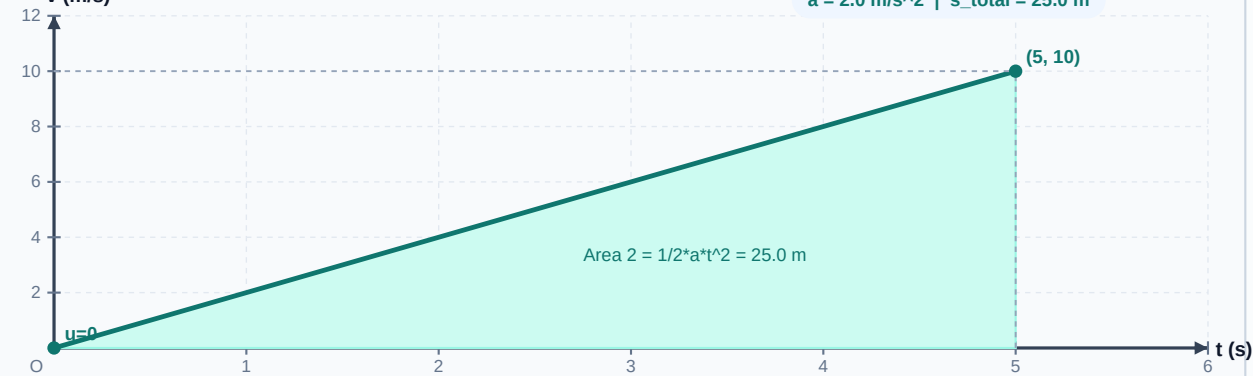
More ways to draw this — Decoder showing the same curve read once as a slope and once as a signed area. Repres

1.12 REPRESENTATION TRANSLATE

Representation Translate

Different pictures of the same idea.

SPEED AGAINST TIME: STEEPNESS AND AREA



More ways to draw this — Velocity-time graph with labelled axes and shaded signed area under the line. Represe

WHEN YOU ARE ALLOWED TO USE THIS RULE

model: phy model graph kinematics

☐ constant positive velocity corresponds to constant positive x-t slope stated / implied / absent

If any condition is absent the relation below this line may not be used.

More ways to check this — Whether the rule is allowed here gate listing each assumption with its stated, impli

CHECKS THAT COULD CATCH A MISTAKE

result under test

- ☐ verify dimensions
- ☐ verify graph slope sign
- ☐ verify signed graph area
- ☐ verify source representation present

Tick only after the check has actually been carried out on the result.

More ways to check this — Verification strip with one tick box per required second, different check. Represent

1.12 REPRESENTATION TRANSLATE

Representation Translate

Different pictures of the same idea.

TWO CASES, ONE THING CHANGED

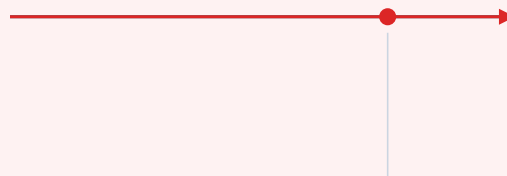
CASE A

the shortcut reproduces the right answer here



CASE B

the shortcut fails here



Decisive feature: A case pair in which the shortcut and the correct model give different answers.

More ways to check this — Two-case contrast panel in which exactly one physical feature differs. Representatio

1.13 STATE PROPAGATE

State Propagate

Picture what is happening before you reach for a formula.

Start with a real situation

A real situation before any symbols.

The object does not teleport: position and velocity at the instant of the change belong to both phases.

(shortened to fit this page)

The big idea

The one sentence worth remembering.

Solve one stage at a time, and hand the end of each stage to the start of the next.

At a phase boundary the state is continuous

(shortened to fit this page)

When does this model work?

Check the situation fits before you use the rule.

Name the model, list its assumptions, and mark each assumption as stated, implied or absent.

acceleration is constant within each phase

acceleration is constant within each stated phase

(shortened to fit this page)

WHAT IS ACTUALLY HAPPENING

- 1 Read what physically happens, in order.
- 2 Mark the start state and the end state.
- 3 Mark the quantity the question asks for.



Link the real situation to a drawing before you calculate.

See it step by step

Draw what is happening before you write anything.

State the origin, the positive direction, and which object the frame is attached to.

Draw a phase strip and mark the quantity that must carry over unchanged.

1. Mark every instant at which the motion changes character.
2. Give each phase its own state table and relation.

(shortened to fit this page)

1.13 STATE PROPAGATE

State Propagate

One question, solved all the way to a number.

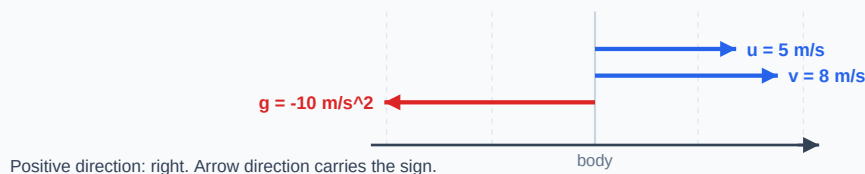
Turn the Picture Into Symbols

Each mark in the drawing gets a name.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

WHICH WAY EACH ARROW POINTS



The same situation drawn out. No new numbers are introduced here.

Watch one worked example

Every step, with the numbers.

A hot-air balloon is rising steadily at 8 m/s when a small sandbag slips off the side. Take gravity as 10 m/s² and ignore air resistance. How long after it slips off does the sandbag stop rising, and how much higher does it get first?

Set it up: you are following the sandbag, from the moment it leaves the balloon, measuring from the point where it slips off, counting upwards as positive.

What you are told: velocity the sandbag has at the moment it slips off = 8 m/s; acceleration due to gravity = 10 m/s². What you want: time until it stops rising.

1. Set Up Your Directions: Call upwards positive, so the sandbag's starting velocity is positive and gravity is negative. Mixing those up is the usual way this goes wrong. Up is positive. Zero is the point where it slips off.
2. See It: Draw two arrows on the sandbag at the moment of release: a velocity arrow pointing up, an acceleration arrow pointing down. They fight each other. Velocity arrow up, acceleration arrow down — so it is still rising but slowing.
3. Which Rule Fits Here?: This is the decisive bit. The sandbag does not start from rest: it leaves the balloon already moving up at the balloon's speed. From then on only gravity acts on it. It keeps the balloon's upward velocity at release, then slows at g until it momentarily stops.

(shortened to fit this page)

STATE PROPAGATE — watch it solved

Watch one worked example (continued)

5. Do the Working: From $v^2 = u^2 - 2gh$ with the final velocity zero, the extra rise is $u^2 / 2g$. $\text{rise} = u^2 / (2g) = 8^2 / (2 \cdot 10) = 3.2 \text{ m}$

6. What the Answer Means: It climbs a little further before it starts to come down, which is why a dropped object from a rising balloon does not head straight for the ground. It keeps rising for a moment, then turns round.

Answer: The sandbag keeps rising for another 0.8 s, climbing a further 3.2 m before it starts to fall.

One thing to check: At the very top the sandbag's speed is zero, but its acceleration is still 10 m/s^2 downwards. Zero speed does not mean zero acceleration — if it did, it would hang there forever.

Check it a different way: Check the velocity at the instant you found. Starting at 8 m/s upwards and losing
(shortened to fit this page)

Easy mistake to make

Lots of people fall for this one.

What lots of people think: Applying a single whole-journey relation across a change of acceleration.

Why that feels right: The shortcut looks sufficient because it reproduces the right answer in the cases seen so far, before the decisive physical feature is tested.

Now compare these two cases: A case pair in which the shortcut and the correct model give different answers.

Fix it like this: Predict both cases with the shortcut.

Fix it like this: Work both cases with the correct model.

Fix it like this: Say in one sentence which feature the shortcut ignores.

(shortened to fit this page)

1.13 STATE PROPAGATE

State Propagate

Check it before you accept it.

Does the answer make sense?

Would this answer make sense in real life?

A check that can never come out wrong is not a check. A real one has to be able to tell you the answer is wrong.

Check you were allowed to use that rule

Verify phase state continuity

(shortened to fit this page)

Quick check

Thirty seconds, no notes.

A hot-air balloon is rising steadily at 10 m/s when a small sandbag slips off the side. Take gravity as 10 m/s² and ignore air resistance. How long after it slips off does the sandbag stop rising, and how much higher does it get first?

Without looking back, say what your first move would be on the last question.

(shortened to fit this page)

Ready to practise?

You will meet it again here.

Where this idea turns up next: the same idea in a situation that looks nothing like this one. You will meet

(shortened to fit this page)

1.13 STATE PROPAGATE

State Propagate

Try it with help, then with less, then on your own.

Try it with help

The first move is done for you.

A hot-air balloon is rising steadily at 5 m/s when a small sandbag slips off the side. Take gravity as 10 m/s^2 and ignore air resistance. How long after it slips off does the sandbag stop rising, and how much higher does it get first?

What you are told: velocity the sandbag has at the moment it slips off = 5 m/s; acceleration due to gravity = 10 m/s^2 . What you want: time until it stops rising.

Use the supplied first-move cue and the pre-drawn frame; complete the remaining steps.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

Try with less help

One of the cues is gone now.

A hot-air balloon is rising steadily at 12 m/s when a small sandbag slips off the side. Take gravity as 10 m/s^2 and ignore air resistance. How long after it slips off does the sandbag stop rising, and how much higher does it get first?

What you are told: velocity the sandbag has at the moment it slips off = 12 m/s; acceleration due to gravity = 10 m/s^2 . What you want: time until it stops rising.

One supplied cue has been removed; decide that step yourself and complete the rest.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

1.13 STATE PROPAGATE

State Propagate

Try it with help, then with less, then on your own.

Now try it yourself

No cues this time. You have everything you need.

A hot-air balloon is rising steadily at 6 m/s when a small sandbag slips off the side. Take gravity as 10 m/s² and ignore air resistance. How long after it slips off does the sandbag stop rising, and how much higher does it get first?

What you are told: velocity the sandbag has at the moment it slips off = 6 m/s; acceleration due to gravity = 10 m/s². What you want: time until it stops rising.

No procedural cue is supplied beyond the situation and the required representation.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

WHERE ZERO IS AND WHICH WAY IS POSITIVE



Declared positive direction: right. Re-sign every given quantity against it here.

More ways to see this — Signed axis strip marking the origin, the declared positive direction, and each signed

WHERE IT IS AT EQUAL TIME STEPS

Gaps are equal time steps; unequal gaps mean the speed is changing.



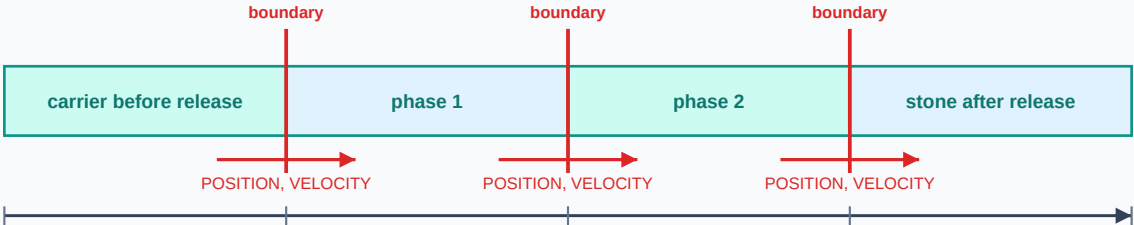
More ways to see this — Strip of successive body positions at equal time intervals along the declared axis. Re

1.13 STATE PROPAGATE

State Propagate

Different pictures of the same idea.

WHAT CARRIES OVER WHEN THE MOTION CHANGES



The end state of one phase is the start state of the next; only the rate may jump.

More ways to draw this — Phase strip with boundary markers and the continuity state labelled across each bound

WHAT YOU KNOW AT THE START AND AT THE END

state variable	stated in the source	to find
acceleration	?	?
displacement	?	?
position	?	?
time	?	?
velocity	8 m/s	?

Fill both instants first; the empty cell is what the question asks for.

More ways to draw this — Ruled state table with a start row and an end row and the unknown cell marked. Repres

WHEN YOU ARE ALLOWED TO USE THIS RULE

model: phy model constant g 1d, phy model moving release, phy model piecewise constant

- | | |
|--|---------------------------|
| ☐ acceleration is constant within each phase | stated / implied / absent |
| ☐ acceleration is constant within each stated phase | stated / implied / absent |
| ☐ air resistance neglected | stated / implied / absent |
| ☐ g constant after release | stated / implied / absent |
| ☐ phase-boundary state is propagated continuously | stated / implied / absent |

If any condition is absent the relation below this line may not be used.

More ways to check this — Whether the rule is allowed here gate listing each assumption with its stated, impli

1.13 STATE PROPAGATE

State Propagate

Different pictures of the same idea.

CHECKS THAT COULD CATCH A MISTAKE

result under test

- ☐ ▶ check you were allowed to use that rule
- ☐ ▶ verify phase state continuity
- ☐ ▶ verify release state continuity
- ☐ ▶ verify source representation present

Tick only after the check has actually been carried out on the result.

More ways to check this — Verification strip with one tick box per required second, different check. Represent

TWO CASES, ONE THING CHANGED

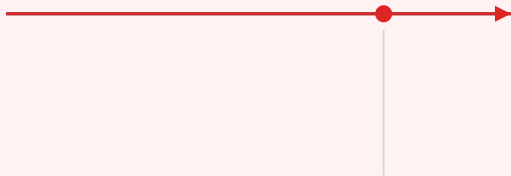
CASE A

the shortcut reproduces the right answer here



CASE B

the shortcut fails here



Decisive feature: A case pair in which the shortcut and the correct model give different answers.

More ways to check this — Two-case contrast panel in which exactly one physical feature differs. Representatio

1.14 UNIFORM ACCELERATION

Uniform Acceleration

Warm up, try it, then check it.

You already know this one

The one sentence worth remembering.

Brief activation: Mark the object and the instants the question refers to.

Now try it yourself

No cues this time. You have everything you need.

A skateboarder starts from rest on a flat straight path and speeds up steadily at 3 m/s^2 . How long does it take her to reach 7.5 m/s ?

(shortened to fit this page)

Does the answer make sense?

Would this answer make sense in real life?

Verify acceleration sign units

Verify constant g assumptions

Verify dimensions

Verify interval not cumulative

(shortened to fit this page)

Quick check

Thirty seconds, no notes.

Say why your answer makes sense in the real situation.

Ready to practise?

You will meet it again here.

Continue without reteaching once the second, different check has passed.

1.14 UNIFORM ACCELERATION

Uniform Acceleration

Different pictures of the same idea.

WHAT EACH LETTER MEANS IN THE PICTURE

DIAGRAM FEATURE



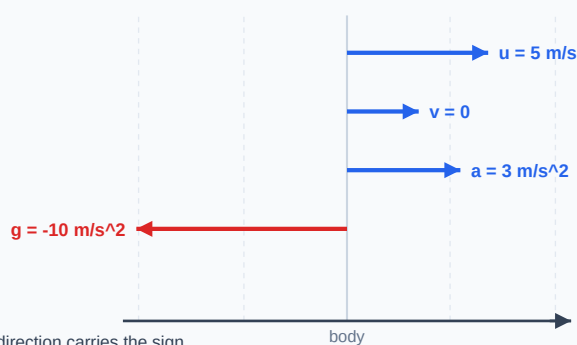
SYMBOL IN THE RELATION

- u** arrow at the start instant
- v** arrow at the end instant
- a** change of the arrow between the instants
- t** span between the two instants

Bind every symbol to a diagram feature before substituting.

More ways to check this — Bridge panel linking each symbol of the relation to the diagram feature it names. Re

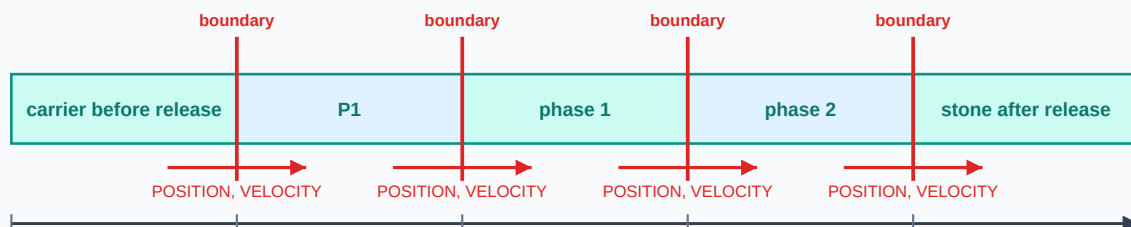
WHICH WAY EACH ARROW POINTS



Positive direction: right. Arrow direction carries the sign.

More ways to draw this — Vector state view with one labelled arrow per directed quantity at the named instant.

WHAT CARRIES OVER WHEN THE MOTION CHANGES



The end state of one phase is the start state of the next; only the rate may jump.

More ways to draw this — Phase strip with boundary markers and the continuity state labelled across each bound

1.14 UNIFORM ACCELERATION

Uniform Acceleration

Different pictures of the same idea.

WHERE ZERO IS AND WHICH WAY IS POSITIVE



Declared positive direction: right. Re-sign every given quantity against it here.

More ways to see this — Signed axis strip marking the origin, the declared positive direction, and each signed

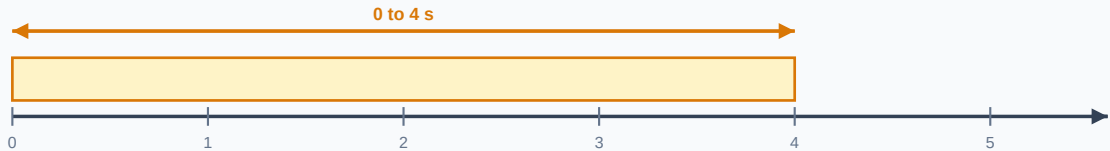
WHAT YOU KNOW AT THE START AND AT THE END

state variable	stated in the source	to find
acceleration	3 m/s ²	?
displacement	?	?
height	45 m	?
position	?	?
time	5 s	?
velocity	0	?

Fill both instants first; the empty cell is what the question asks for.

More ways to draw this — Ruled state table with a start row and an end row and the unknown cell marked. Repres

ONE NAMED SECOND IS A DIFFERENCE



Compute the quantity at each bounding instant, then subtract.

More ways to draw this — Timeline marking the bounding instants of the named interval and the span between the

1.14 UNIFORM ACCELERATION

Uniform Acceleration

Different pictures of the same idea.

CHECKS THAT COULD CATCH A MISTAKE

result under test

- ☐ ▶ verify acceleration sign units
- ☐ ▶ verify constant g assumptions
- ☐ ▶ verify dimensions
- ☐ ▶ verify interval not cumulative
- ☐ ▶ verify maximum height endpoint

Tick only after the check has actually been carried out on the result.

More ways to check this — Verification strip with one tick box per required second, different check. Represent

1.15 VERTICAL PROJECTION

Vertical Projection

Picture what is happening before you reach for a formula.

Start with a real situation

A real situation before any symbols.

Ask what is being assumed constant, and whether the situation says it is.

(shortened to fit this page)

The big idea

The one sentence worth remembering.

Before using a relation, say in words what it assumes and point to where the question says that is true.

A relation is only valid while its assumptions
(shortened to fit this page)

When does this model work?

Check the situation fits before you use the rule.

Name the model, list its assumptions, and mark each assumption as stated, implied or absent.

air resistance neglected
(shortened to fit this page)

WHAT IS ACTUALLY HAPPENING

- 1 Read what physically happens, in order.
- 2 Mark the start state and the end state.
- 3 Mark the quantity the question asks for.



Link the real situation to a drawing before you calculate.

See it step by step

Draw what is happening before you write anything.

State the origin, the positive direction, and which object the frame is attached to.

Check each assumption of the candidate model against a stated condition.

1. List the candidate physical models for this situation.

2. Write each model's assumptions as a checklist.

(shortened to fit this page)

1.15 VERTICAL PROJECTION

Vertical Projection

One question, solved all the way to a number.

Turn the Picture Into Symbols

Each mark in the drawing gets a name.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

WHAT YOU KNOW AT THE START AND AT THE END

state variable	stated in the source	to find
acceleration	?	?
height	?	?

Fill both instants first; the empty cell is what the question asks for.

The same situation drawn out. No new numbers are introduced here.

Watch one worked example

Every step, with the numbers.

A ball is thrown straight up from the ground at 20 m/s. Take gravity as 10 m/s² and ignore air resistance. How high does the ball get, and how long does it take to get there?

Set it up: you are following the ball, measuring from the ground, where it leaves the hand, counting upwards as positive.

What you are told: speed it is thrown at = 20 m/s; acceleration due to gravity = 10 m/s². What you want: greatest height reached.

1. Set Up Your Directions: Up is positive, so the throw is positive and gravity is negative. That one decision fixes every sign after it. Up is positive. Zero is the ground.
2. See It: Draw the ball at three moments: leaving the hand, halfway up, and at the top. The velocity arrow shrinks; the gravity arrow never does. The velocity arrow shrinks to nothing at the top; the gravity arrow stays the same all the way.
3. Which Rule Fits Here?: With air resistance ignored, the acceleration is steady all the way up, so the constant-acceleration relations hold. The key extra fact is that at the highest point the velocity is momentarily zero. Steady acceleration g downwards, and velocity zero at the top.
4. Do the Working: It loses g of upward velocity every second, so it runs out after u/g seconds. $t_{up} = u/g = 20/10 = 2\text{ s}$

(shortened to fit this page)

VERTICAL PROJECTION — watch it solved

Watch one worked example (continued)

squared over $2g$. This relation skips the time, which we were not asked for first. $h_{\text{max}} = u^2/(2g) = 20^2/(2 \times 10) = 20 \text{ m}$

6. What the Answer Means: Throwing it twice as fast makes it go four times as high, because the height depends on the speed squared. Double the throwing speed, quadruple the height.

Answer: The ball reaches 20 m, taking 2 s to get there.

One thing to check: At the top the ball's speed is zero, but gravity has not switched off — the acceleration is still 10 m/s^2 downwards. That is why it does not stay up there.

Check it a different way: Use the time instead of the speeds. Over the trip to the top, the height is u times t (shortened to fit this page)

Easy mistake to make

Lots of people fall for this one.

What lots of people think: Using a constant-rate relation for a situation that changes rate part-way through.

Why that feels right: The shortcut looks sufficient because it reproduces the right answer in the cases seen so far, before the decisive physical feature is tested.

Now compare these two cases: A case pair in which the shortcut and the correct model give different answers.

Fix it like this: Predict both cases with the shortcut.

Fix it like this: Work both cases with the correct model.

Fix it like this: Say in one sentence which feature the shortcut ignores.

(shortened to fit this page)

1.15 VERTICAL PROJECTION

Vertical Projection

Check it before you accept it.

Does the answer make sense?

Would this answer make sense in real life?

A check that can never come out wrong is not a check. A real one has to be able to tell you the answer is wrong.

Verify constant g assumptions

Verify dimensions

(shortened to fit this page)

Quick check

Thirty seconds, no notes.

A ball is thrown straight up from the ground at 12 m/s. Take gravity as 10 m/s^2 and ignore air resistance. How high does the ball get, and how long does it take to get there?

Without looking back, say what your first move would be on the last question.

(shortened to fit this page)

Ready to practise?

You will meet it again here.

Where this idea turns up next: the same idea in a situation that looks nothing like this one. You will meet

(shortened to fit this page)

Try it with help

The first move is done for you.

A ball is thrown straight up from the ground at 15 m/s. Take gravity as 10 m/s^2 and ignore air resistance. How high does the ball get, and how long does it take to get there?

What you are told: speed it is thrown at = 15 m/s; acceleration due to gravity = 10 m/s^2 . What you want: greatest height reached.

Use the supplied first-move cue and the pre-drawn frame; complete the remaining steps.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

1.15 VERTICAL PROJECTION

Vertical Projection

Try it with help, then with less, then on your own.

Try with less help

One of the cues is gone now.

A ball is thrown straight up from the ground at 25 m/s. Take gravity as 10 m/s^2 and ignore air resistance. How high does the ball get, and how long does it take to get there?

What you are told: speed it is thrown at = 25 m/s; acceleration due to gravity = 10 m/s^2 . What you want: greatest height reached.

One supplied cue has been removed; decide that step yourself and complete the rest.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

Now try it yourself

No cues this time. You have everything you need.

A ball is thrown straight up from the ground at 30 m/s. Take gravity as 10 m/s^2 and ignore air resistance. How high does the ball get, and how long does it take to get there?

What you are told: speed it is thrown at = 30 m/s; acceleration due to gravity = 10 m/s^2 . What you want: greatest height reached.

No procedural cue is supplied beyond the situation and the required representation.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

1.15 VERTICAL PROJECTION

Vertical Projection

Different pictures of the same idea.

WHERE ZERO IS AND WHICH WAY IS POSITIVE



Declared positive direction: upward. Re-sign every given quantity against it here.

More ways to see this — Signed axis strip marking the origin, the declared positive direction, and each signed

WHAT EACH LETTER MEANS IN THE PICTURE

DIAGRAM FEATURE



SYMBOL IN THE RELATION

u arrow at the start instant

Bind every symbol to a diagram feature before substituting.

More ways to check this — Bridge panel linking each symbol of the relation to the diagram feature it names. Re

WHEN YOU ARE ALLOWED TO USE THIS RULE

model: phy model constant g 1d

- | | |
|---|---------------------------|
| ☐ air resistance neglected | stated / implied / absent |
| ☐ g constant | stated / implied / absent |

If any condition is absent the relation below this line may not be used.

More ways to check this — Whether the rule is allowed here gate listing each assumption with its stated, impli

1.15 VERTICAL PROJECTION

Vertical Projection

Different pictures of the same idea.

CHECKS THAT COULD CATCH A MISTAKE

result under test

- ☐ verify constant g assumptions
- ☐ verify dimensions
- ☐ verify maximum height endpoint

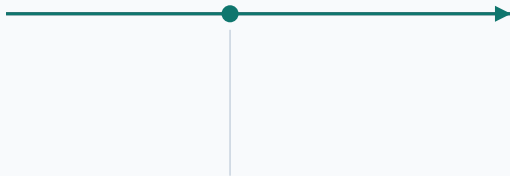
Tick only after the check has actually been carried out on the result.

More ways to check this — Verification strip with one tick box per required second, different check. Represent

TWO CASES, ONE THING CHANGED

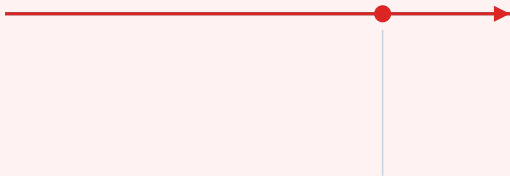
CASE A

the shortcut reproduces the right answer here



CASE B

the shortcut fails here



Decisive feature: A case pair in which the shortcut and the correct model give different answers.

More ways to check this — Two-case contrast panel in which exactly one physical feature differs. Representatio

1.16 VT SIGNED AREA

Vt Signed Area

Picture what is happening before you reach for a formula.

Start with a real situation

A real situation before any symbols.

A graph is a record of the motion, not a picture of the path.

(shortened to fit this page)

The big idea

The one sentence worth remembering.

Ask first: do I need how fast this is changing, or how much it has changed in total?

On a graph of a state variable against time, slope is a rate and signed area is a change; which one

(shortened to fit this page)

When does this model work?

Check the situation fits before you use the rule.

Slope gives the rate of the plotted quantity;

(shortened to fit this page)

WHAT IS ACTUALLY HAPPENING

- 1 Read what physically happens, in order.
- 2 Mark the start state and the end state.
- 3 Mark the quantity the question asks for.



Link the real situation to a drawing before you calculate.

See it step by step

Draw what is happening before you write anything.

State the origin, the positive direction, and which object the frame is attached to.

Decode slope and area separately and say what each one is physically.

1. Read and state both axis labels and units.
2. Decide whether the question asks for a rate or an accumulated change.

(shortened to fit this page)

1.16 VT SIGNED AREA

Vt Signed Area

One question, solved all the way to a number.

Turn the Picture Into Symbols

Each mark in the drawing gets a name.

State the origin, the positive direction, and which object the frame is attached to.

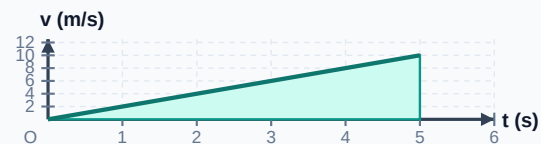
(shortened to fit this page)

STEEPNESS OR AREA? ONE GRAPH, TWO READINGS

READ AS SLOPE = a rate



READ AS SIGNED AREA = a change



The same situation drawn out. No new numbers are introduced here.

Watch one worked example

Every step, with the numbers.

The velocity-time graph of a trolley is a straight line starting at 0 m/s and reaching 12 m/s after 4 s. How far does the trolley travel in that time?

Set it up: you are following the trolley, measuring from where it is when the clock starts, counting the direction the graph calls positive as positive.

What you are told: velocity at the end of the interval = 12 m/s; length of the interval = 4 s. What you want: distance travelled.

1. Set Up Your Directions: Read the axes before anything else. Velocity up, time across — so areas on this graph have metres per second times seconds, which is metres. Up axis: velocity. Across axis: time. Area therefore means distance.
2. See It: Shade the region between the line and the time axis. Here it is a triangle, because the line starts at zero. A shaded triangle: base is the time, height is the final velocity.
3. Which Rule Fits Here?: The question asks how far, not how fast it is changing — so this is an area question, not a steepness question. That is the decision that matters on any velocity-time graph. Area under a velocity-time graph is the distance travelled.
4. Do the Working: The area of a triangle is half the base times the height. $s = 0.5 \cdot v_{\text{end}} \cdot T = 0.5 \cdot 12 \cdot 4 =$

(shortened to fit this page)

VT SIGNED AREA — watch it solved

Watch one worked example (continued)

5. What the Answer Means: A positive area above the axis means it moved forwards that far. That is how far forwards it got in the interval.

6. Check Your Answer: It must be less than it would have travelled at the top speed the whole way, because it spent most of the time slower than that. Less than the top speed times the time.

Answer: The trolley travels 24 m in that time.

One thing to check: Area under a velocity-time graph comes out in metres. If your answer is in m/s you read the height of the graph instead of the area under it.

Check it a different way: Use the average velocity instead of the area. It starts at 0 and finishes at 12 m/s,
(shortened to fit this page)

Easy mistake to make

Lots of people fall for this one.

What lots of people think: Reading a velocity-time graph as if it were a picture of the path travelled.

Why that feels right: The shortcut looks sufficient because it reproduces the right answer in the cases seen so far, before the decisive physical feature is tested.

Now compare these two cases: A case pair in which the shortcut and the correct model give different answers.

Fix it like this: Predict both cases with the shortcut.

Fix it like this: Work both cases with the correct model.

Fix it like this: Say in one sentence which feature the shortcut ignores.

(shortened to fit this page)

1.16 VT SIGNED AREA

Vt Signed Area

Check it before you accept it.

Does the answer make sense?

Would this answer make sense in real life?

A check that can never come out wrong is not a check. A real one has to be able to tell you the answer is wrong.

Verify dimensions

Verify signed graph area

(shortened to fit this page)

Quick check

Thirty seconds, no notes.

The velocity-time graph of a trolley is a straight line starting at 0 m/s and reaching 6 m/s after 4 s. How far does the trolley travel in that time?

Without looking back, say what your first move would be on the last question.

(shortened to fit this page)

Ready to practise?

You will meet it again here.

Where this idea turns up next: the same idea in a situation that looks nothing like this one. You will meet

(shortened to fit this page)

Try it with help

The first move is done for you.

The velocity-time graph of a trolley is a straight line starting at 0 m/s and reaching 10 m/s after 5 s. How far does the trolley travel in that time?

What you are told: velocity at the end of the interval = 10 m/s; length of the interval = 5 s. What you want: distance travelled.

Use the supplied first-move cue and the pre-drawn frame; complete the remaining steps.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

1.16 VT SIGNED AREA

Vt Signed Area

Try it with help, then with less, then on your own.

Try with less help

One of the cues is gone now.

The velocity-time graph of a trolley is a straight line starting at 0 m/s and reaching 8 m/s after 2 s. How far does the trolley travel in that time?

What you are told: velocity at the end of the interval = 8 m/s; length of the interval = 2 s. What you want: distance travelled.

One supplied cue has been removed; decide that step yourself and complete the rest.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

Now try it yourself

No cues this time. You have everything you need.

The velocity-time graph of a trolley is a straight line starting at 0 m/s and reaching 16 m/s after 6 s. How far does the trolley travel in that time?

What you are told: velocity at the end of the interval = 16 m/s; length of the interval = 6 s. What you want: distance travelled.

No procedural cue is supplied beyond the situation and the required representation.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

1.16 VT SIGNED AREA

Vt Signed Area

Different pictures of the same idea.

SPEED AGAINST TIME: STEEPNESS AND AREA



More ways to draw this — Velocity-time graph with labelled axes and shaded signed area under the line. Represe

CHECKS THAT COULD CATCH A MISTAKE

result under test

- ☐ verify dimensions
- ☐ verify signed graph area
- ☐ verify source representation present

Tick only after the check has actually been carried out on the result.

More ways to check this — Verification strip with one tick box per required second, different check. Represent

TWO CASES, ONE THING CHANGED

CASE A

the shortcut reproduces the right answer her
e



CASE B

the shortcut fails here



Decisive feature: A case pair in which the shortcut and the correct model give different answers.

More ways to check this — Two-case contrast panel in which exactly one physical feature differs. Representatio

1.17 XT SLOPE VELOCITY

Xt Slope Velocity

Picture what is happening before you reach for a formula.

Start with a real situation

A real situation before any symbols.

A graph is a record of the motion, not a picture of the path.

(shortened to fit this page)

The big idea

The one sentence worth remembering.

Ask first: do I need how fast this is changing, or how much it has changed in total?

On a graph of a state variable against time, slope is a rate and signed area is a change; which one

(shortened to fit this page)

When does this model work?

Check the situation fits before you use the rule.

Name the model, list its assumptions, and mark each assumption as stated, implied or absent.

(shortened to fit this page)

WHAT IS ACTUALLY HAPPENING

- 1 Read what physically happens, in order.
- 2 Mark the start state and the end state.
- 3 Mark the quantity the question asks for.



Link the real situation to a drawing before you calculate.

See it step by step

Draw what is happening before you write anything.

State the origin, the positive direction, and which object the frame is attached to.

Decode slope and area separately and say what each one is physically.

1. Read and state both axis labels and units.
2. Decide whether the question asks for a rate or an accumulated change.

(shortened to fit this page)

1.17 XT SLOPE VELOCITY

Xt Slope Velocity

One question, solved all the way to a number.

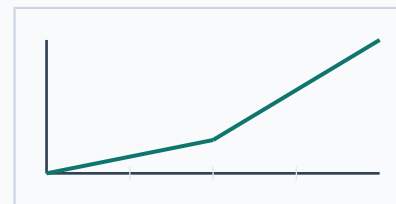
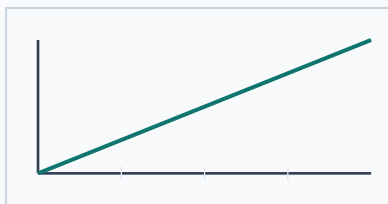
Turn the Picture Into Symbols

Each mark in the drawing gets a name.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

THE CHOICES, DRAWN ON THE SAME AXES



Reject each option by naming the feature it gets wrong.

The same situation drawn out. No new numbers are introduced here.

Watch one worked example

Every step, with the numbers.

The position-time graph of a robot on a straight rail is a straight line. At 0 s the robot is at 0 m, and at 5 s it is at 10 m. How fast is the robot moving?

Set it up: you are following the robot, measuring from the zero mark on the rail, counting the direction the position axis counts up as positive.

What you are told: position at the first instant = 0 m; first instant = 0 s; position at the second instant = 10 m; second instant = 5 s. What you want: velocity.

1. Set Up Your Directions: The up axis here is position, not velocity. That one difference changes what the steepness means, so read the axes first. Up axis: position in m. Across axis: time in s.
2. See It: Mark both instants on the line and draw the step between them: across for the time passed, up for the ground covered. A step under the line: across is the time, up is the distance covered.
3. Which Rule Fits Here?: A straight position-time line means the robot covers the same ground every second, so its velocity never changes. Straight position-time line means constant velocity, and the steepness is that velocity.
4. Do the Working: Steepness is change in position divided by change in time — which is exactly what velocity means. $v = (x_2 - x_1)/(t_2 - t_1) = (10 - 0)/(5 - 0) = 2 \text{ m/s}$

(shortened to fit this page)

XT SLOPE VELOCITY — watch it solved

Watch one worked example (continued)

It covers that much rail every second, always in the same direction.

6. Check Your Answer: Metres divided by seconds is metres per second, which is a velocity. The units come out as m/s.

Answer: The robot is moving at 2 m/s, and because the line is straight it keeps that velocity the whole time.

One thing to check: On a position-time graph the steepness is the velocity, and the height is only where the robot is. A high flat line means far away and not moving, not fast.

Check it a different way: Run it forwards instead. If that really is the velocity, then over the same stretch of
(shortened to fit this page)

Easy mistake to make

Lots of people fall for this one.

What lots of people think: Reading a velocity-time graph as if it were a picture of the path travelled.

Why that feels right: The shortcut looks sufficient because it reproduces the right answer in the cases seen so far, before the decisive physical feature is tested.

Now compare these two cases: A case pair in which the shortcut and the correct model give different answers.

Fix it like this: Predict both cases with the shortcut.

Fix it like this: Work both cases with the correct model.

Fix it like this: Say in one sentence which feature the shortcut ignores.

(shortened to fit this page)

1.17 XT SLOPE VELOCITY

Xt Slope Velocity

Check it before you accept it.

Does the answer make sense?

Would this answer make sense in real life?

A check that can never come out wrong is not a check. A real one has to be able to tell you the answer is wrong.

Verify graph slope sign

(shortened to fit this page)

Quick check

Thirty seconds, no notes.

The position-time graph of a robot on a straight rail is a straight line. At 1 s the robot is at 2 m, and at 10 s it is at 10 m. How fast is the robot moving?

Without looking back, say what your first move would be on the last question.

(shortened to fit this page)

Ready to practise?

You will meet it again here.

Where this idea turns up next: the same idea in a situation that looks nothing like this one. You will meet

(shortened to fit this page)

Try it with help

The first move is done for you.

The position-time graph of a robot on a straight rail is a straight line. At 1 s the robot is at 2 m, and at 4 s it is at 20 m. How fast is the robot moving?

What you are told: position at the first instant = 2 m; first instant = 1 s; position at the second instant = 20 m; second instant = 4 s. What you want: velocity.

Use the supplied first-move cue and the pre-drawn frame; complete the remaining steps.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

1.17 XT SLOPE VELOCITY

Xt Slope Velocity

Try it with help, then with less, then on your own.

Try with less help

One of the cues is gone now.

The position-time graph of a robot on a straight rail is a straight line. At 2 s the robot is at 5 m, and at 6 s it is at 14 m. How fast is the robot moving?

What you are told: position at the first instant = 5 m; first instant = 2 s; position at the second instant = 14 m; second instant = 6 s. What you want: velocity.

One supplied cue has been removed; decide that step yourself and complete the rest.

The full answer is in the answers appendix — try it first.

(shortened to fit this page)

Now try it yourself

No cues this time. You have everything you need.

The position-time graph of a robot on a straight rail is a straight line. At 0 s the robot is at 0 m, and at 8 s it is at 18 m. How fast is the robot moving?

What you are told: position at the first instant = 0 m; first instant = 0 s; position at the second instant = 18 m; second instant = 8 s. What you want: velocity.

No procedural cue is supplied beyond the situation and the required representation.

The full answer is in the answers appendix — try it first.

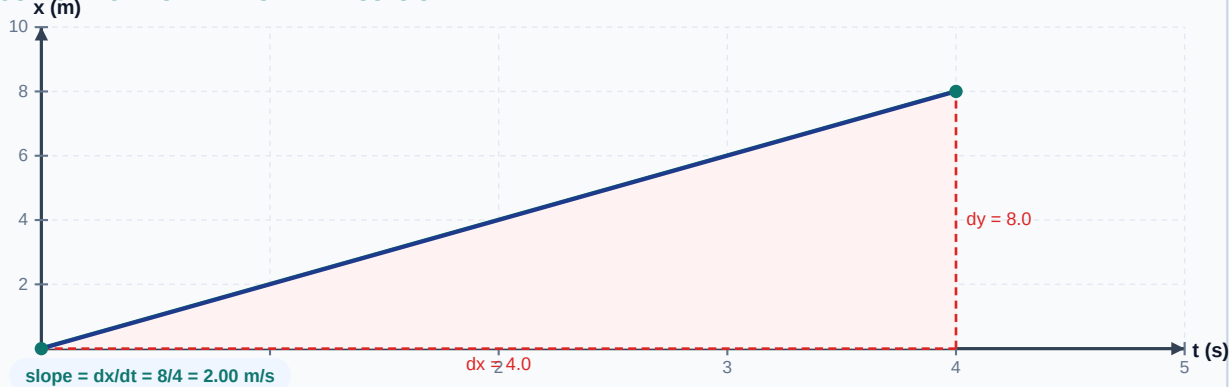
(shortened to fit this page)

1.17 XT SLOPE VELOCITY

Xt Slope Velocity

Different pictures of the same idea.

POSITION AGAINST TIME: STEEPNESS IS SPEED



More ways to draw this — Position-time graph with labelled axes and a rise-over-run slope triangle. Representa

WHEN YOU ARE ALLOWED TO USE THIS RULE

model: phy model graph kinematics

☐ constant positive velocity corresponds to constant positive x-t slope stated / implied / absent

If any condition is absent the relation below this line may not be used.

More ways to check this — Whether the rule is allowed here gate listing each assumption with its stated, impli

CHECKS THAT COULD CATCH A MISTAKE

result under test

- ☐ verify graph slope sign
- ☐ verify source representation present

Tick only after the check has actually been carried out on the result.

More ways to check this — Verification strip with one tick box per required second, different check. Represent

1.17 XT SLOPE VELOCITY

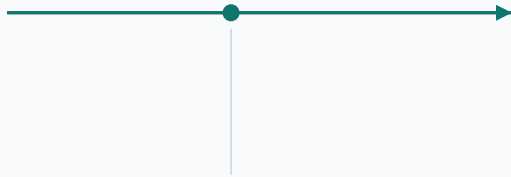
Xt Slope Velocity

Different pictures of the same idea.

TWO CASES, ONE THING CHANGED

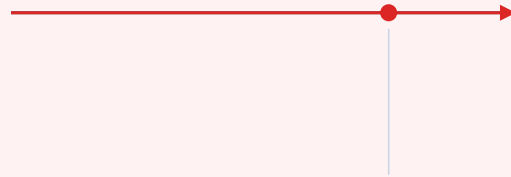
CASE A

the shortcut reproduces the right answer here



CASE B

the shortcut fails here



Decisive feature: A case pair in which the shortcut and the correct model give different answers.

More ways to check this — Two-case contrast panel in which exactly one physical feature differs. Representatio

A Appendix A — Core Practice

PRACTICE

Work these first. The answers are in the next appendix.

Question 1 — Try It With Help

A cyclist rides along a straight road. She covers the first half of the road at a steady 30 km/h and the
(shortened to fit this page)

Question 2 — Try It With Less Help

A cyclist rides along a straight road. She covers the first half of the road at a steady 20 km/h and the
(shortened to fit this page)

Question 3 — Try It Yourself

A cyclist rides along a straight road. She covers the first half of the road at a steady 60 km/h and the
(shortened to fit this page)

Question 4 — Try It With Help

A runner starts at point A on a semicircular track of radius 35 m and runs along the curve all the way round
(shortened to fit this page)

Question 5 — Try It With Less Help

A runner starts at point A on a semicircular track of radius 50 m and runs along the curve all the way round
(shortened to fit this page)

Question 6 — Try It Yourself

A runner starts at point A on a semicircular track of radius 14 m and runs along the curve all the way round
(shortened to fit this page)

Question 7 — Try It With Help

A skateboarder starts from rest on a flat straight path and speeds up steadily at 2.5 m/s^2 . How long does it take to reach 10 m/s ?
(shortened to fit this page)

Question 8 — Try It With Less Help

A skateboarder starts from rest on a flat straight path and speeds up steadily at 3 m/s^2 . How long does it take to reach 10 m/s ?
(shortened to fit this page)

Question 9 — Try It Yourself

A skateboarder starts from rest on a flat straight path and speeds up steadily at 1.5 m/s^2 . How long does it take to reach 10 m/s ?
(shortened to fit this page)

Question 10 — Try It With Help

A skateboarder starts from rest on a flat straight path and speeds up steadily at 2.5 m/s^2 . How long does it take to reach 10 m/s ?
(shortened to fit this page)

Question 11 — Try It With Less Help

A skateboarder starts from rest on a flat straight path and speeds up steadily at 3 m/s^2 . How long does it take to reach 10 m/s ?
(shortened to fit this page)

Question 12 — Try It Yourself

A skateboarder starts from rest on a flat straight path and speeds up steadily at 1.5 m/s^2 . How long does it take to reach 10 m/s ?
(shortened to fit this page)

Question 13 — Try It With Help

A small stone is dropped — not thrown — from a bridge 20 m above the water. Take gravity as 10 m/s^2
(shortened to fit this page)

Question 14 — Try It With Less Help

A small stone is dropped — not thrown — from a bridge 80 m above the water. Take gravity as 10 m/s^2
(shortened to fit this page)

Question 15 — Try It Yourself

A small stone is dropped — not thrown — from a bridge 125 m above the water. Take gravity as 10 m/s^2
(shortened to fit this page)

Question 16 — Try It With Help

A velocity-time graph for a trolley is a straight line. At 0 s the line reads 0 m/s, and at 6 s it reads 12 m/s.
(shortened to fit this page)

Question 17 — Try It With Less Help

A velocity-time graph for a trolley is a straight line. At 1 s the line reads 4 m/s, and at 8 s it reads 10 m/s.

(shortened to fit this page)

Question 18 — Try It Yourself

A velocity-time graph for a trolley is a straight line. At 2 s the line reads 1 m/s, and at 4 s it reads 9 m/s.

(shortened to fit this page)

Question 19 — Try It With Help

A skateboarder starts from rest on a flat straight path and speeds up steadily at 2.5 m/s^2 . How long does

(shortened to fit this page)

Question 20 — Try It With Less Help

A skateboarder starts from rest on a flat straight path and speeds up steadily at 3 m/s^2 . How long does it

(shortened to fit this page)

Question 21 — Try It Yourself

A skateboarder starts from rest on a flat straight path and speeds up steadily at 1.5 m/s^2 . How long does
(shortened to fit this page)

Question 22 — Try It With Help

A hot-air balloon is rising steadily at 5 m/s when a small sandbag slips off the side. Take gravity as 10
(shortened to fit this page)

Question 23 — Try It With Less Help

A hot-air balloon is rising steadily at 12 m/s when a small sandbag slips off the side. Take gravity as 10
(shortened to fit this page)

Question 24 — Try It Yourself

A hot-air balloon is rising steadily at 6 m/s when a small sandbag slips off the side. Take gravity as 10
(shortened to fit this page)

Question 25 — Try It With Help

A cart is already rolling at 4 m/s along a straight track. For the first 4 s it speeds up steadily at 3 m/s^2 .

(shortened to fit this page)

Question 26 — Try It With Less Help

A cart is already rolling at 6 m/s along a straight track. For the first 2 s it speeds up steadily at 2.5 m/s^2 .

(shortened to fit this page)

Question 27 — Try It Yourself

A cart is already rolling at 2 m/s along a straight track. For the first 5 s it speeds up steadily at 2 m/s^2 .

(shortened to fit this page)

Question 28 — Try It With Help

A go-kart is moving at 0 m/s and then speeds up steadily at 4 m/s^2 . How far does it travel during the 6th

(shortened to fit this page)

Question 29 — Try It With Less Help

A go-kart is moving at 3 m/s and then speeds up steadily at 1.5 m/s^2 . How far does it travel during the
(shortened to fit this page)

Question 30 — Try It Yourself

A go-kart is moving at 1 m/s and then speeds up steadily at 3 m/s^2 . How far does it travel during the 5th
(shortened to fit this page)

Question 31 — Try It With Help

Two stones are released from rest at the same instant, one from 40 m up and the other from 35 m up.
(shortened to fit this page)

Question 32 — Try It With Less Help

Two stones are released from rest at the same instant, one from 60 m up and the other from 50 m up.
(shortened to fit this page)

Question 33 — Try It Yourself

Two stones are released from rest at the same instant, one from 80 m up and the other from 20 m up.

(shortened to fit this page)

Question 34 — Try It With Help

A velocity-time graph for a trolley is a straight line. At 0 s the line reads 0 m/s, and at 6 s it reads 12 m/s.

(shortened to fit this page)

Question 35 — Try It With Less Help

A velocity-time graph for a trolley is a straight line. At 1 s the line reads 4 m/s, and at 8 s it reads 10 m/s.

(shortened to fit this page)

Question 36 — Try It Yourself

A velocity-time graph for a trolley is a straight line. At 2 s the line reads 1 m/s, and at 4 s it reads 9 m/s.

(shortened to fit this page)

Question 37 — Try It With Help

A hot-air balloon is rising steadily at 5 m/s when a small sandbag slips off the side. Take gravity as 10
(shortened to fit this page)

Question 38 — Try It With Less Help

A hot-air balloon is rising steadily at 12 m/s when a small sandbag slips off the side. Take gravity as 10
(shortened to fit this page)

Question 39 — Try It Yourself

A hot-air balloon is rising steadily at 6 m/s when a small sandbag slips off the side. Take gravity as 10
(shortened to fit this page)

Question 40 — Try It Yourself

A skateboarder starts from rest on a flat straight path and speeds up steadily at 1.5 m/s^2 . How long does
(shortened to fit this page)

Question 41 — Try It With Help

A ball is thrown straight up from the ground at 15 m/s. Take gravity as 10 m/s^2 and ignore air resistance.

(shortened to fit this page)

Question 42 — Try It With Less Help

A ball is thrown straight up from the ground at 25 m/s. Take gravity as 10 m/s^2 and ignore air resistance.

(shortened to fit this page)

Question 43 — Try It Yourself

A ball is thrown straight up from the ground at 30 m/s. Take gravity as 10 m/s^2 and ignore air resistance.

(shortened to fit this page)

Question 44 — Try It With Help

The velocity-time graph of a trolley is a straight line starting at 0 m/s and reaching 10 m/s after 2 s. How

(shortened to fit this page)

Question 45 — Try It With Less Help

The velocity-time graph of a trolley is a straight line starting at 0 m/s and reaching 8 m/s after 6 s. How far
(shortened to fit this page)

Question 46 — Try It Yourself

The velocity-time graph of a trolley is a straight line starting at 0 m/s and reaching 16 m/s after 4 s. How
(shortened to fit this page)

Question 47 — Try It With Help

The position-time graph of a robot on a straight rail is a straight line. At 0 s the robot is at 0 m, and at 4 s it
(shortened to fit this page)

Question 48 — Try It With Less Help

The position-time graph of a robot on a straight rail is a straight line. At 1 s the robot is at 2 m, and at 6 s it
(shortened to fit this page)

Question 49 — Try It Yourself

The position-time graph of a robot on a straight rail is a straight line. At 2 s the robot is at 5 m, and at 8 s it
(shortened to fit this page)

B Appendix B — Core Solutions

ANSWERS

Compare these with your own working.

Answer 1

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 2

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 3

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 4

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. See It: Build the required representation and check it against the state table.

*(shortened to fit this page)***Answer 5**

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. See It: Build the required representation and check it against the state table.

*(shortened to fit this page)***Answer 6**

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. See It: Build the required representation and check it against the state table.

(shortened to fit this page)

Answer 7

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 8

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 9

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 10

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 11

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 12

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 13

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 14

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 15

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 16

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. See It: Build the required representation and check it against the state table.

*(shortened to fit this page)***Answer 17**

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. See It: Build the required representation and check it against the state table.

*(shortened to fit this page)***Answer 18**

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. See It: Build the required representation and check it against the state table.

(shortened to fit this page)

Answer 19

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 20

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 21

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 22

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 23

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 24

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 25

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 26

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 27

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 28

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 29

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 30

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 31

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 32

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 33

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 34

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. See It: Build the required representation and check it against the state table.

*(shortened to fit this page)***Answer 35**

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. See It: Build the required representation and check it against the state table.

*(shortened to fit this page)***Answer 36**

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. See It: Build the required representation and check it against the state table.

(shortened to fit this page)

Answer 37

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 38

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 39

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 40

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 41

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 42

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 43

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. Select model: Name the physical model and check each of its assumptions against a stated condition.
(shortened to fit this page)

Answer 44

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. See It: Build the required representation and check it against the state table.
(shortened to fit this page)

Answer 45

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. See It: Build the required representation and check it against the state table.
(shortened to fit this page)

Answer 46

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. See It: Build the required representation and check it against the state table.

*(shortened to fit this page)***Answer 47**

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. See It: Build the required representation and check it against the state table.

*(shortened to fit this page)***Answer 48**

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. See It: Build the required representation and check it against the state table.

(shortened to fit this page)

Answer 49

1. Read phenomenon: Read the described situation and say what physically happens, in order.
2. Define system: Name the object being tracked and declare the origin and positive direction.
3. Extract knowns and hidden facts: Enter every given quantity into the start and end state rows, including facts implied by the wording.
4. See It: Build the required representation and check it against the state table.

(shortened to fit this page)

C Appendix C — Printable Handout

PRINTABLE

A one-page reminder of how to start. No answers here.

AVERAGE SPEED EQUAL DISTANCE

First move: Mark the object and the instants the question refers to.

This skill declares no frame or sign obligation; state the system anyway before writing symbols.

(shortened to fit this page)

DISTANCE DISPLACEMENT

First move: Mark the object and the instants the question refers to.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

FIRST EQUATION

First move: Mark the object and the instants the question refers to.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

FRAME SIGN SETUP

First move: Mark the object and the instants the question refers to.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

FREE FALL CONSTANT G

First move: Mark the object and the instants the question refers to.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

GRAPH ACCELERATION

First move: Mark the object and the instants the question refers to.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

Checking a rule fits the situation

First move: Mark the object and the instants the question refers to.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

MOVING RELEASE

First move: Mark the object and the instants the question refers to.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

MULTIPHASE KINEMATICS

First move: Mark the object and the instants the question refers to.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

NTH SECOND INTERVAL

First move: Mark the object and the instants the question refers to.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

RELATIVE ACCELERATION SAME G

First move: Mark the object and the instants the question refers to.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

REPRESENTATION TRANSLATE

First move: Mark the object and the instants the question refers to.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

STATE PROPAGATE

First move: Mark the object and the instants the question refers to.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

UNIFORM ACCELERATION

Brief activation: Mark the object and the instants the question refers to.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

VERTICAL PROJECTION

First move: Mark the object and the instants the question refers to.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

VT SIGNED AREA

First move: Mark the object and the instants the question refers to.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)

XT SLOPE VELOCITY

First move: Mark the object and the instants the question refers to.

State the origin, the positive direction, and which object the frame is attached to.

(shortened to fit this page)